```
1116  0  INSERT FILE='E:/Documents/GitHub/biostatistics-portfolio/_spss_tmp/assignment-2-
cox-regression.sps' ERROR=CONTINUE.
1116  0  * Set the working directory to the folder that contains this repository, then ru
n.
1117  0  CD 'E:/Documents/GitHub/biostatistics-portfolio'.
1118  0  * Encoding: UTF-8.
1119  0
1120  0
1121  0  GET DATA
1122  0    /TYPE=XLSX
1123  0    /FILE='data/raw/survival_assignment_Number_II_2026.xlsx'
1124  0    /SHEET=name 'Sheet1'
1125  0    /CELLRANGE=FULL
1126  0    /READNAMES=ON
1127  0    /DATATYPEMIN PERCENTAGE=95.0
1128  0    /HIDDEN IGNORE=YES.
1129  0  EXECUTE.
1130  0  DATASET NAME DataSet1 WINDOW=FRONT.
1131  0
1132  0  *Comment: imports the Excel sheet (206 rows).
1133  0
1134  0  DELETE VARIABLES V9 V10 V11 V12.
1135  0  RENAME VARIABLES (Column1 Column2 Column3 Column4 Column5 Column6 Column7 Column
8 =
1136  0    ID Year_surgery Age Status Survtime Surg_care Heart_transplant Wait_time).
1137  0
1138  0
1139  0  VARIABLE LABELS
1140  0    ID 'Patient identification number'
1141  0    Year_surgery 'Year when accepted surgery'
1142  0    Age 'Patient''s age'
1143  0    Status 'Status'
1144  0    Survtime 'Survival time in days'
1145  0    Surg_care 'Received surgical care'
1146  0    Heart_transplant 'Received heart transplant'
1147  0    Wait_time 'Waiting time before surgery'.
1148  0
1149  0
1150  0
1151  0  VALUE LABELS Status 0 'censored' 1 'died'
1152  0    /Surg_care 0 'No' 1 'Yes'
1153  0    /Heart_transplant 0 'No' 1 'Yes'.
1154  0
1155  0  *Comment: Status 1 = died (the event), 0 = censored. Surgical care and transplan
t are coded 1 = Yes, 0 = No.
1156  0
1157  0  VARIABLE LEVEL ID Status Surg_care Heart_transplant (NOMINAL)
1158  0    /Year_surgery Age Survtime Wait_time (SCALE).
1159  0
```

```
1160  0  *Comment: categorical variables set to nominal, numeric variables to scale.
1161  0
1162  0  FORMATS Survtime (F6.1) Wait_time (F3.0).
1163  0
1164  0
1165  0
1166  0  *inspect categories
1167  0
1168  0  FREQUENCIES VARIABLES=Status Surg_care Heart_transplant.
```

Frequencies

Notes

Output Created		03-NOV-2026 15:35:47
Comments		
Input	Active Dataset	DataSet1
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	206
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics are based on all cases with valid data.
Syntax		FREQUENCIES VARIABLES=Status Surg_care Heart_transplant.
Resources	Processor Time	00:00:00.06
	Elapsed Time	00:00:00.01

Statistics

		Status	Received surgical care	Received heart transplant
N	Valid	206	206	206
	Missing	0	0	0

Frequency Table

Status

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	censored	56	27.2	27.2	27.2
	died	150	72.8	72.8	100.0
	Total	206	100.0	100.0	

Received surgical care

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	No	182	88.3	88.3	88.3
	Yes	24	11.7	11.7	100.0
	Total	206	100.0	100.0	

Received heart transplant

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	No	68	33.0	33.0	33.0
	Yes	138	67.0	67.0	100.0
	Total	206	100.0	100.0	

```
1169  0
1170  0  *Comment: 150 died and 56 censored. 24 rows had surgical care and 138 had a hear
t transplant.
1171  0  *No unexpected codes. Verdict: categories clean.
1172  0
1173  0  *inspect continous variables
1174  0
1175  0  DATASET ACTIVATE DataSet1.
1176  0  DESCRIPTIVES VARIABLES=Year_surgery Age Survtime Wait_time
1177  0    /STATISTICS=MEAN STDDEV MIN MAX.
```

Descriptives

Notes

Output Created		03-NOV-2026 15:35:48
Comments		
Input	Active Dataset	DataSet1
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	206
Missing Value Handling	Definition of Missing	User defined missing values are treated as missing.
	Cases Used	All non-missing data are used.
Syntax		DESCRIPTIVES VARIABLES=Year_surgery Age Survtime Wait_time /STATISTICS=MEAN STDDEV MIN MAX.
Resources	Processor Time	00:00:00.02
	Elapsed Time	00:00:00.02

Descriptive Statistics

	N	Minimum	Maximum	Mean	Std. Deviation
Year when accepted surgery	206	67	74	70.62	1.895
Patient's age	206	8	64	44.64	9.774
Survival time in days	206	1.0	1799.0	310.079	426.9224
Waiting time before surgery	206	0	310	25.74	44.648
Valid N (listwise)	206				

```
1178  0
1179  0  *Comment: no implausible values. Survtime runs from 1 to 1799 days.
1180  0  *Ages include two young outliers (8 and 19), plausible and kept. Verdict: contin
uous variables clean.
1181  0
1182  0  *check missing ness; none seen according to variable view;
1183  0
1184  0  *check duplicates
1185  0
1186  0
1187  0
```

```
1188  0  * Identify Duplicate Cases.
1189  0  SORT CASES BY ID(A).
1190  0  MATCH FILES
1191  0    /FILE=*
1192  0    /BY ID
1193  0    /FIRST=PrimaryFirst
1194  0    /LAST=Primaryidlast.
1195  0  DO IF (PrimaryFirst).
1196  1  COMPUTE  MatchSequence=1-Primaryidlast.
1197  1  ELSE.
1198  1  COMPUTE  MatchSequence=MatchSequence+1.
1199  1  END IF.
1200  0  LEAVE  MatchSequence.
1201  0  FORMATS  MatchSequence (f7).
1202  0  COMPUTE  InDupGrp=MatchSequence>0.
1203  0  SORT CASES InDupGrp(D).
1204  0  MATCH FILES
1205  0    /FILE=*
1206  0    /DROP=PrimaryFirst InDupGrp MatchSequence.
1207  0  VARIABLE LABELS  Primaryidlast 'Indicator of each last matching case as Primary'
.
1208  0  VALUE LABELS  Primaryidlast 0 'Duplicate Case' 1 'Primary Case'.
1209  0  VARIABLE LEVEL  Primaryidlast (ORDINAL).
1210  0  FREQUENCIES VARIABLES=Primaryidlast.
```

Frequencies

Notes

Output Created		03-NOV-2026 15:35:48
Comments		
Input	Active Dataset	DataSet1
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	206
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics are based on all cases with valid data.
Syntax		FREQUENCIES VARIABLES=Primaryidlast.
Resources	Processor Time	00:00:00.02
	Elapsed Time	00:00:00.01

Statistics

Indicator of each last matching case as Primary

N	Valid	206
	Missing	0

Indicator of each last matching case as Primary

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Primary Case	206	100.0	100.0	100.0

```
1211  0  EXECUTE.
1212  0
1213  0  *Comment: checking on ID alone finds no repeated IDs (every row is a primary cas
e),
1214  0  *so it does not detect the duplication. Verdict: no duplicate IDs.
1215  0
1216  0
1217  0  *duplicates without ID
1218  0
1219  0
1220  0  SORT CASES BY Year_surgery(A) Age(A) Status(A) Survtime(A) Surg_care(A) Heart_tr
ansplant(A)
1221  0      Wait_time(A).
1222  0  MATCH FILES
1223  0    /FILE=*
1224  0    /BY Year_surgery Age Status Survtime Surg_care Heart_transplant Wait_time
1225  0    /FIRST=PrimaryFirst
1226  0    /LAST=Primarynoid.
1227  0  DO IF (PrimaryFirst).
1228  1  COMPUTE  MatchSequence=1-Primarynoid.
1229  1  ELSE.
1230  1  COMPUTE  MatchSequence=MatchSequence+1.
1231  1  END IF.
1232  0  LEAVE  MatchSequence.
1233  0  FORMATS  MatchSequence (f7).
1234  0  COMPUTE  InDupGrp=MatchSequence>0.
1235  0  SORT CASES InDupGrp(D).
1236  0  MATCH FILES
1237  0    /FILE=*
1238  0    /DROP=PrimaryFirst InDupGrp MatchSequence.
1239  0  VARIABLE LABELS  Primarynoid 'Indicator of each last matching case as Primary'.
1240  0  VALUE LABELS  Primarynoid 0 'Duplicate Case' 1 'Primary Case'.
1241  0  VARIABLE LEVEL  Primarynoid (ORDINAL).
1242  0  FREQUENCIES VARIABLES=Primarynoid.
```

Frequencies

Notes

Output Created		03-NOV-2026 15:35:48
Comments		
Input	Active Dataset	DataSet1
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	206
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics are based on all cases with valid data.
Syntax		FREQUENCIES VARIABLES=Primarynoid.
Resources	Processor Time	00:00:00.00
	Elapsed Time	00:00:00.01

Statistics

Indicator of each last matching case as Primary

N	Valid	206
	Missing	0

Indicator of each last matching case as Primary

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Duplicate Case	103	50.0	50.0	50.0
	Primary Case	103	50.0	50.0	100.0
	Total	206	100.0	100.0	

```
1243  0  EXECUTE.
1244  0
1245  0
1246  0  *Comment: Matching on all variables except ID shows 103 of the 206 records (50%)
 are exact duplicates of another record. Th
           e dataset contains 103 unique patients, each entered twice under different ID
 numbers. An ID-only check does not detect
           this.
1247  0
1248  0  *Fit the model
1249  0
```

```
1250  0
1251  0
1252  0  COXREG Survtime
1253  0    /STATUS=Status(1)
1254  0    /PATTERN BY Heart_transplant
1255  0    /CONTRAST (Surg_care)=Indicator(1)
1256  0    /CONTRAST (Heart_transplant)=Indicator(1)
1257  0    /METHOD=ENTER Year_surgery Age Surg_care Heart_transplant Wait_time
1258  0    /PLOT LML
1259  0    /SAVE=LML PRESID DFBETA
1260  0    /PRINT=CI(95)
1261  0    /CRITERIA=PIN(.05) POUT(.10) ITERATE(20).
```

Cox Regression

Notes

Output Created		03-NOV-2026 15:35:49
Comments		
Input	Active Dataset	DataSet1
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	206
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
Syntax		COXREG Survtime /STATUS=Status(1) /PATTERN BY Heart_transplant /CONTRAST (Surg_care) =Indicator(1) /CONTRAST (Heart_transplant) =Indicator(1) /METHOD=ENTER Year_surgery Age Surg_care Heart_transplant Wait_time /PLOT LML /SAVE=LML PRESID DFBETA /PRINT=CI(95) /CRITERIA=PIN(.05) POUT(.10) ITERATE(20).

Notes

Resources	Processor Time	00:00:01.49
	Elapsed Time	00:00:01.32
Variables Created or Modified	LML_1	Log-minus-log-of-survival function
	PR1_1	Partial residual for Year when accepted surgery
	PR2_1	Partial residual for Patient's age
	PR3_1	Partial residual for Received surgical care
	PR4_1	Partial residual for Received heart transplant
	PR5_1	Partial residual for Waiting time before surgery
	DFB1_1	Dfbeta for Year when accepted surgery
	DFB2_1	Dfbeta for Patient's age
	DFB3_1	Dfbeta for Received surgical care
	DFB4_1	Dfbeta for Received heart transplant
	DFB5_1	Dfbeta for Waiting time before surgery

Case Processing Summary

		N	Percent
Cases available in analysis	Event[a]	150	72.8%
	Censored	56	27.2%
	Total	206	100.0%
Cases dropped	Cases with missing values	0	0.0%
	Cases with negative time	0	0.0%
	Censored cases before the earliest event in a stratum	0	0.0%
	Total	0	0.0%
Total		206	100.0%

a. Dependent Variable: Survival time in days

Categorical Variable Codings[a,c]

		Frequency	(1)
Received surgical care[b]	0=No	182	0
	1=Yes	24	1
Received heart transplant[b]	0=No	68	0
	1=Yes	138	1

a. Category variable: Received surgical care (Surg_care)

b. Indicator Parameter Coding

c. Category variable: Received heart transplant (Heart_transplant)

Block 0: Beginning Block

Omnibus Tests of Model Coefficients

-2 Log Likelihood
1401.205

Block 1: Method = Enter

Omnibus Tests of Model Coefficients[a]

-2 Log Likelihood	Overall (score)			Change From Previous Step			Change ...
	Chi-square	df	Sig.	Chi-square	df	Sig.	Chi-square
1296.399	112.667	5	.000	104.806	5	.000	104.806

Omnibus Tests of Model Coefficients[a]

Change From Previous ..	
df	Sig.
5	.000

a. Beginning Block Number 1. Method = Enter

Variables in the Equation

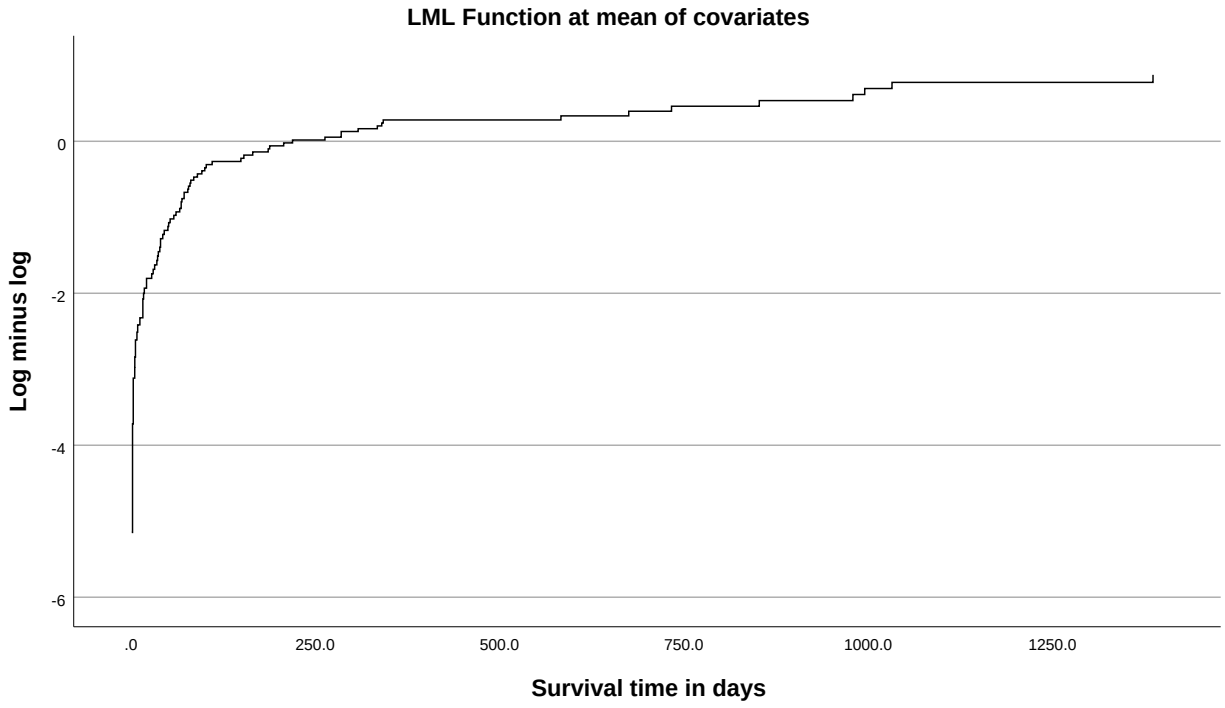
	B	SE	Wald	df	Sig.	Exp(B)
Year when accepted surgery	-.077	.047	2.665	1	.103	.926
Patient's age	.057	.010	30.603	1	.000	1.058
Received surgical care	-.622	.318	3.829	1	.050	.537
Received heart transplant	-1.384	.223	38.406	1	.000	.251
Waiting time before surgery	-.007	.004	4.369	1	.037	.993

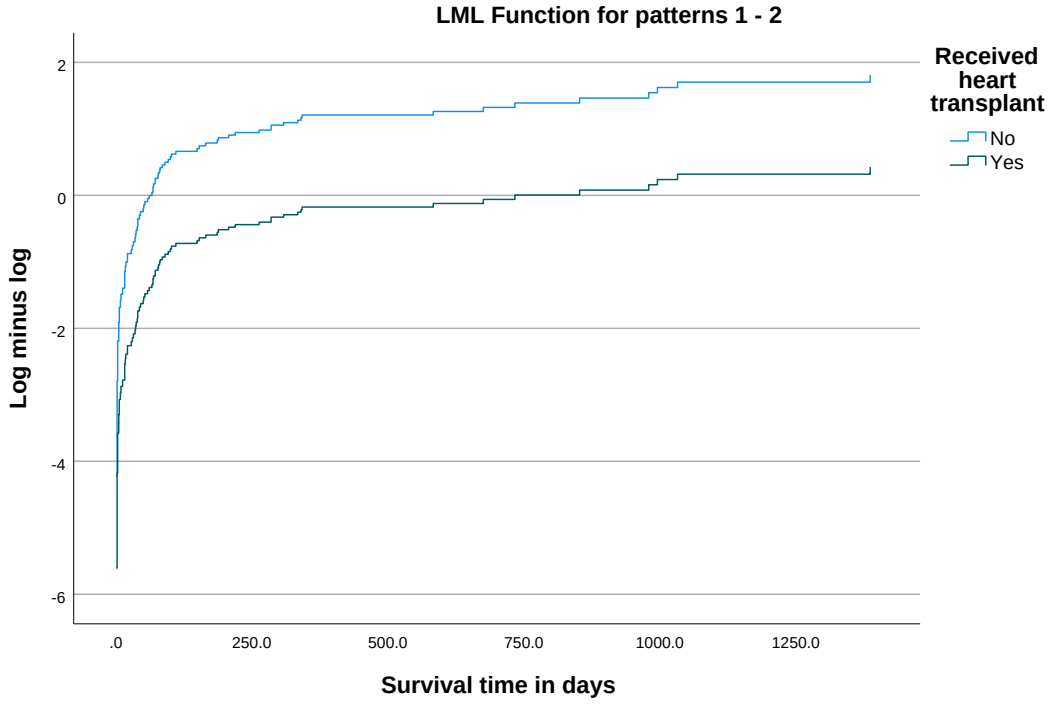
Variables in the Equation

	95.0% CI for Exp(B)	
	Lower	Upper
Year when accepted surgery	.845	1.015
Patient's age	1.037	1.080
Received surgical care	.288	1.001
Received heart transplant	.162	.388
Waiting time before surgery	.986	1.000

Covariate Means and Pattern Values

		Pattern	
	Mean	1	2
Year when accepted surgery	70.621	70.621	70.621
Patient's age	44.641	44.641	44.641
Received surgical care	.117	.117	.117
Received heart transplant	.670	.000	1.000
Waiting time before surgery	25.738	25.738	25.738





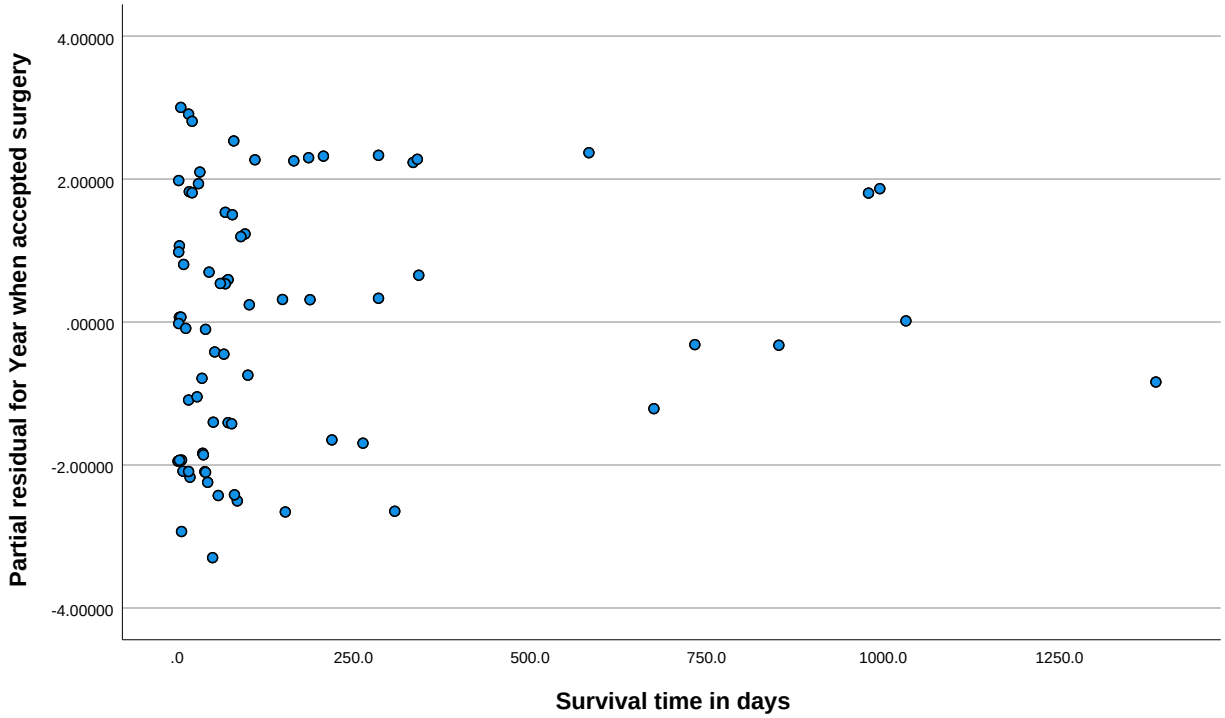

```
1262  0
1263  0  *Comment: baseline Cox model on all 206 rows (150 events, 56 censored, none drop
ped).
```

```
1264  0  *Hazard ratios: year of surgery .926, age 1.058 per year, surgical care .537 (bo
rderline, p = .050),
1265  0  *heart transplant .251, waiting time .993. Reference category is No for both Yes
/No variables.
1266  0  *P-values are too small because of the duplicated rows, and transplant is coded
fixed 0/1,
1267  0  *so immortal-time bias is possible. Verdict: provisional model; proportional haz
ards still to be checked.
1268  0
1269  0  * plots ; Partial residual for Year when accepted surgery vs survtime  (PR1_1)
1270  0
1271  0
1272  0  GRAPH
1273  0    /SCATTERPLOT(BIVAR)=Survtime WITH PR1_1
1274  0    /MISSING=LISTWISE.
```

Graph

Notes

Output Created		03-NOV-2026 15:35:50
Comments		
Input	Active Dataset	DataSet1
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	206
Syntax		GRAPH /SCATTERPLOT(BIVAR) =Survtime WITH PR1_1 /MISSING=LISTWISE.
Resources	Processor Time	00:00:01.17
	Elapsed Time	00:00:01.49

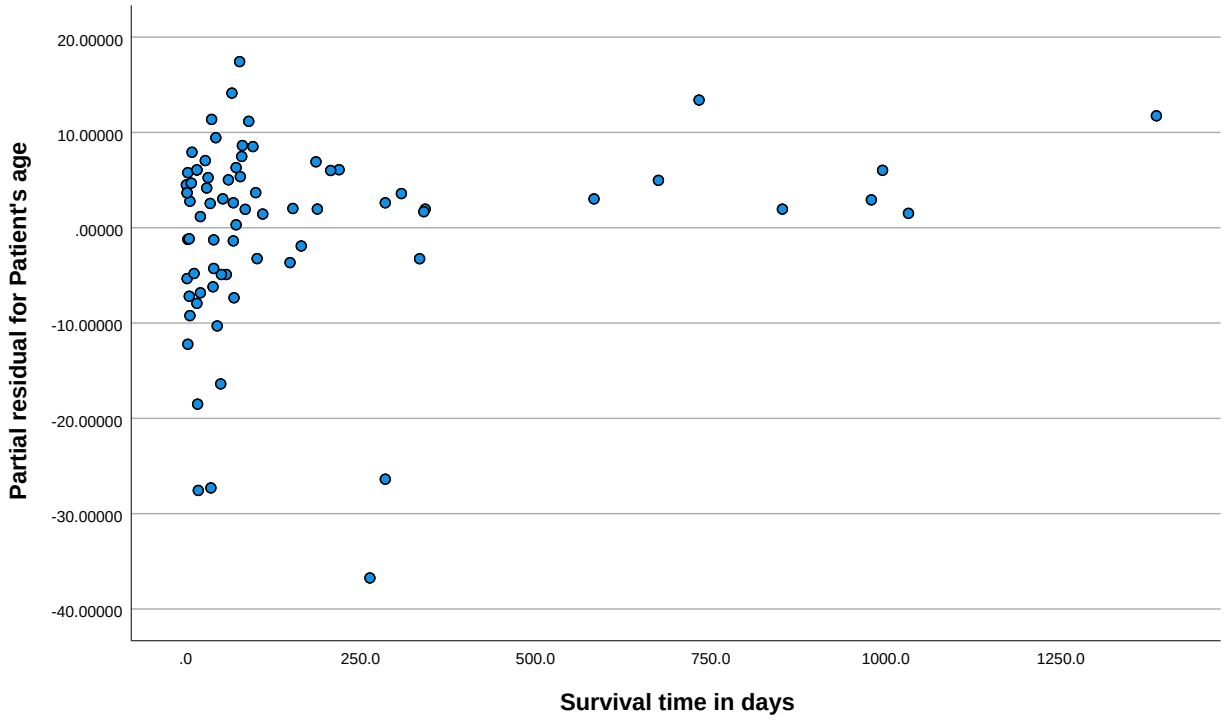



```
1275  0
1276  0  *Partial residuals scatter around zero across follow-up, with no upward or downw
ard slope. The narrowing at later times ref
           lects fewer deaths. Verdict: no trend.
1277  0
1278  0
1279  0  * plots ; Partial residual for Patient's age vs survtime
1280  0
1281  0  GRAPH
1282  0    /SCATTERPLOT(BIVAR)=Survtime WITH PR2_1
1283  0    /MISSING=LISTWISE.
```

Graph

Notes

Output Created		03-NOV-2026 15:35:51
Comments		
Input	Active Dataset	DataSet1
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	206
Syntax		GRAPH /SCATTERPLOT(BIVAR) =Survtime WITH PR2_1 /MISSING=LISTWISE.
Resources	Processor Time	00:00:00.62
	Elapsed Time	00:00:01.29



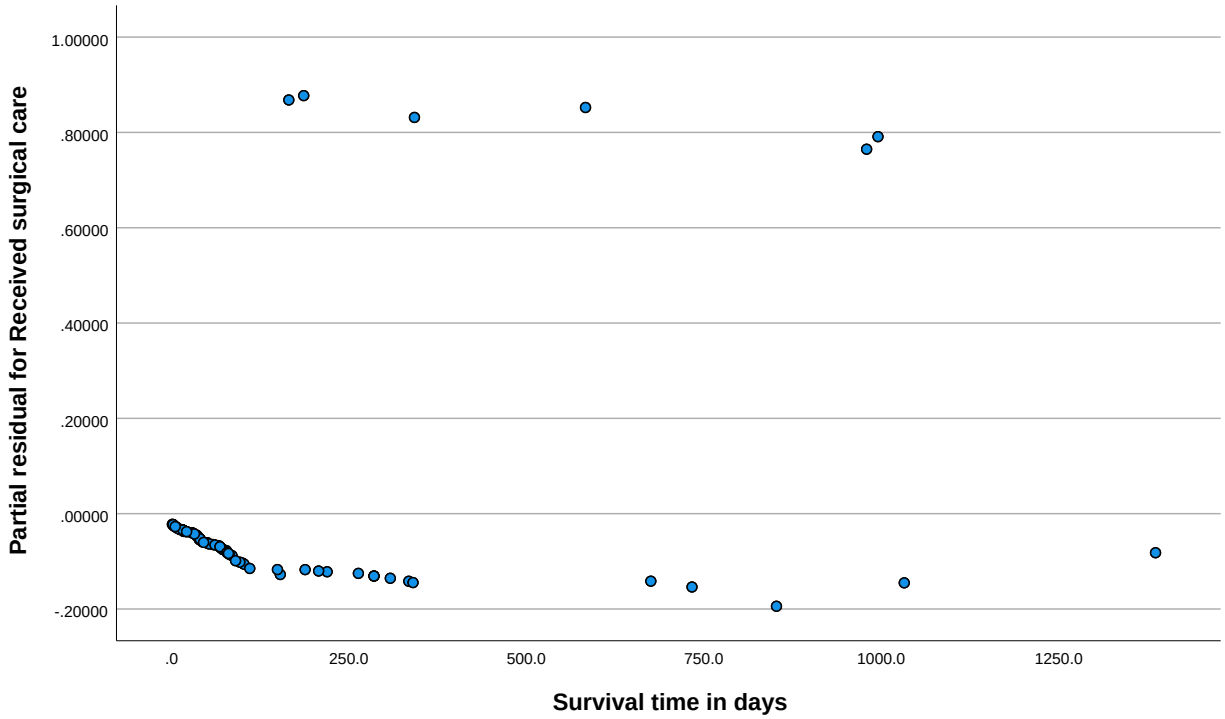

```
1284  0
1285  0  *Early residuals are widely spread, partly from the youngest patients (ages 8 an
d 19). Later residuals sit mostly above zer
           o but rest on few points. No clear slope. Verdict: no clear trend; confirm wi
th correlation test.
1286  0
1287  0
```

```
1288  0  * plots ; Partial residual for Received surgical care VS SURV TIME
1289  0
1290  0  GRAPH
1291  0    /SCATTERPLOT(BIVAR)=Survtime WITH PR3_1
1292  0    /MISSING=LISTWISE.
```

Graph

Notes

Output Created		03-NOV-2026 15:35:53
Comments		
Input	Active Dataset	DataSet1
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	206
Syntax		GRAPH /SCATTERPLOT(BIVAR) =Survtime WITH PR3_1 /MISSING=LISTWISE.
Resources	Processor Time	00:00:01.33
	Elapsed Time	00:00:01.04

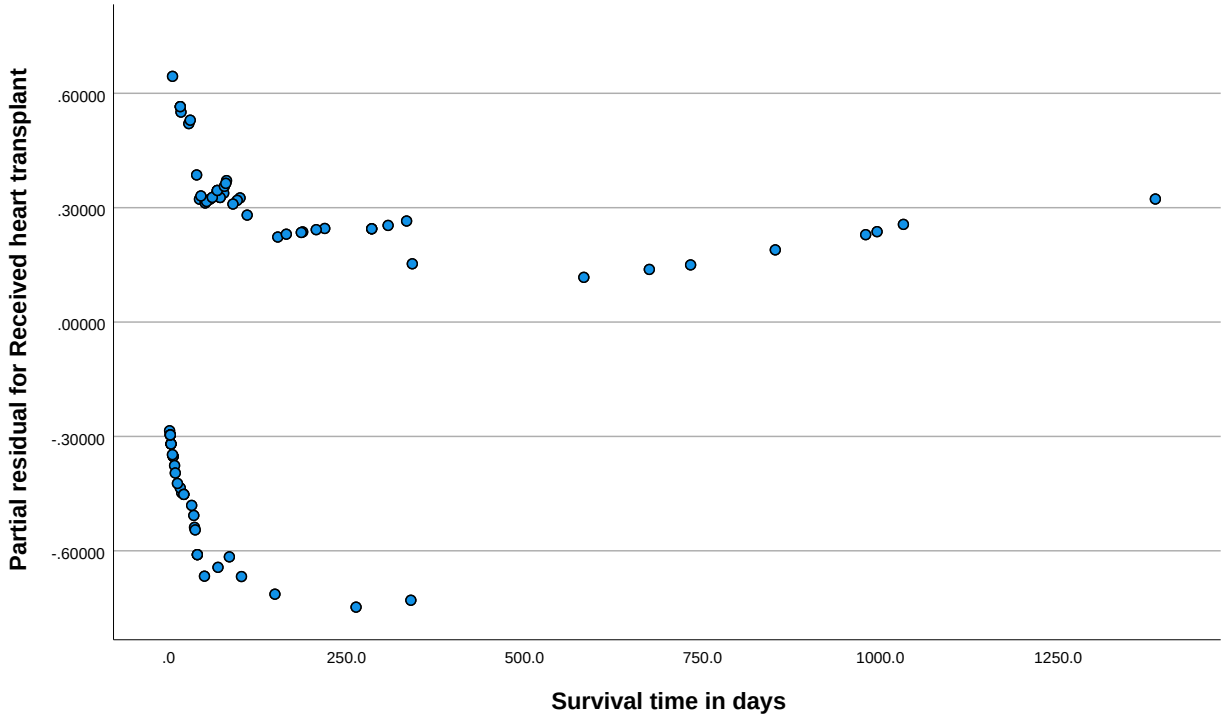


```
1293  0
1294  0  *Residuals form two bands (one per category), so a trend is hard to judge by eye
. Only six deaths occurred in the surgery g
            roup. Verdict: inconclusive.
1295  0
1296  0  * plots ; Partial residual for Received heart transplant VS SURV TIME
1297  0
1298  0  GRAPH
1299  0    /SCATTERPLOT(BIVAR)=Survtime WITH PR4_1
1300  0    /MISSING=LISTWISE.
```

Graph

Notes

Output Created		03-NOV-2026 15:35:54
Comments		
Input	Active Dataset	DataSet1
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	206
Syntax		GRAPH /SCATTERPLOT(BIVAR) =Survtime WITH PR4_1 /MISSING=LISTWISE.
Resources	Processor Time	00:00:00.95
	Elapsed Time	00:00:00.61

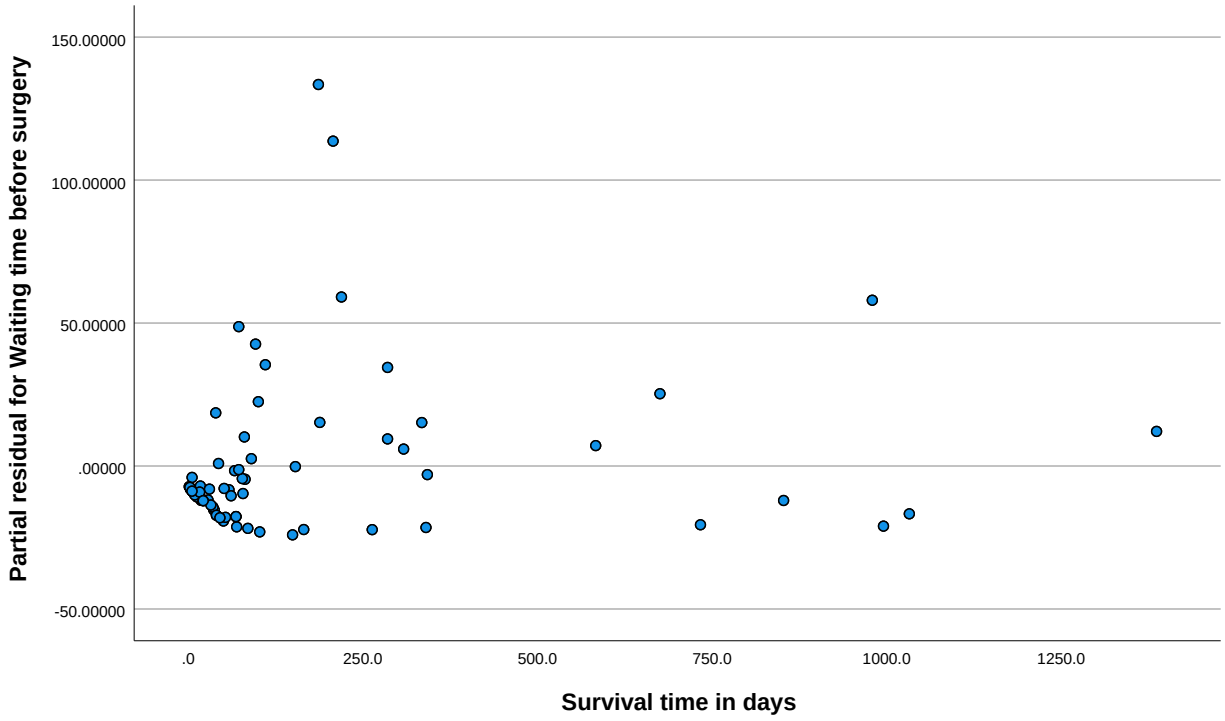



```
1301  0
1302  0  *The two bands converge over the first months, and after about day 340 all death
s are in transplanted patients. This may re
           flect fixed transplant coding. Verdict: trend present; possible violation, co
nfirm with correlation test.
1303  0
1304  0  * plots ; Partial residual for Waiting time before surgery VS SURV TIME
1305  0
1306  0  GRAPH
1307  0    /SCATTERPLOT(BIVAR)=Survtime WITH PR5_1
1308  0    /MISSING=LISTWISE.
```

Graph

Notes

Output Created		03-NOV-2026 15:35:55
Comments		
Input	Active Dataset	DataSet1
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	206
Syntax		GRAPH /SCATTERPLOT(BIVAR) =Survtime WITH PR5_1 /MISSING=LISTWISE.
Resources	Processor Time	00:00:00.48
	Elapsed Time	00:00:00.50




```
1309  0
1310  0  *Most residuals are negative, with a few large positives from long waits (139 an
d 160 days). The drift mirrors the transpla
           nt plot, because waiting time is 0 without a transplant. Verdict: inconclusiv
e.
1311  0
1312  0
```

```
1313  0  *correlation test on the partial residuals
1314  0
1315  0
1316  0
1317  0  CORRELATIONS
1318  0    /VARIABLES=Survtime PR1_1 PR2_1 PR3_1 PR4_1 PR5_1
1319  0    /PRINT=TWOTAIL NOSIG FULL
1320  0    /MISSING=PAIRWISE.
```

Correlations

Notes

Output Created		03-NOV-2026 15:35:55
Comments		
Input	Active Dataset	DataSet1
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	206
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics for each pair of variables are based on all the cases with valid data for that pair.
Syntax		CORRELATIONS /VARIABLES=Survtime PR1_1 PR2_1 PR3_1 PR4_1 PR5_1 /PRINT=TWOTAIL NOSIG FULL /MISSING=PAIRWISE.
Resources	Processor Time	00:00:00.11
	Elapsed Time	00:00:00.09

Correlations

		Survival time in days	Partial residual for Year when accepted surgery	Partial residual for Patient's age
Survival time in days	Pearson Correlation	1	.113	.177*
	Sig. (2-tailed)		.169	.031
	N	206	150	150
Partial residual for Year when accepted surgery	Pearson Correlation	.113	1	.049
	Sig. (2-tailed)	.169		.552
	N	150	150	150
Partial residual for Patient's age	Pearson Correlation	.177*	.049	1
	Sig. (2-tailed)	.031	.552	
	N	150	150	150
Partial residual for Received surgical care	Pearson Correlation	.283**	.307**	.080
	Sig. (2-tailed)	.000	.000	.328
	N	150	150	150
Partial residual for Received heart transplant	Pearson Correlation	.214**	.166*	.307**
	Sig. (2-tailed)	.009	.043	.000
	N	150	150	150
Partial residual for Waiting time before surgery	Pearson Correlation	.154	.209*	.194*
	Sig. (2-tailed)	.059	.010	.017
	N	150	150	150

Correlations

		Partial residual for Received surgical care	Partial residual for Received heart transplant	Partial residual for Waiting time before surgery
Survival time in days	Pearson Correlation	.283**	.214**	.154
	Sig. (2-tailed)	.000	.009	.059
	N	150	150	150
Partial residual for Year when accepted surgery	Pearson Correlation	.307**	.166*	.209*
	Sig. (2-tailed)	.000	.043	.010
	N	150	150	150
Partial residual for Patient's age	Pearson Correlation	.080	.307**	.194*
	Sig. (2-tailed)	.328	.000	.017
	N	150	150	150
Partial residual for Received surgical care	Pearson Correlation	1	.107	.233**
	Sig. (2-tailed)		.194	.004
	N	150	150	150
Partial residual for Received heart transplant	Pearson Correlation	.107	1	.357**
	Sig. (2-tailed)	.194		.000
	N	150	150	150
Partial residual for Waiting time before surgery	Pearson Correlation	.233**	.357**	1
	Sig. (2-tailed)	.004	.000	
	N	150	150	150

*. Correlation is significant at the 0.05 level (2-tailed).

**. Correlation is significant at the 0.01 level (2-tailed).

```
1321  0  NONPAR CORR
1322  0    /VARIABLES=Survtime PR1_1 PR2_1 PR3_1 PR4_1 PR5_1
1323  0    /PRINT=SPEARMAN TWOTAIL NOSIG FULL
1324  0    /MISSING=PAIRWISE.
```

Nonparametric Correlations

Notes

Output Created		03-NOV-2026 15:35:55
Comments		
Input	Active Dataset	DataSet1
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	206
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics for each pair of variables are based on all the cases with valid data for that pair.
Syntax		NONPAR CORR /VARIABLES=Survtime PR1_1 PR2_1 PR3_1 PR4_1 PR5_1 /PRINT=SPEARMAN TWOTAIL NOSIG FULL /MISSING=PAIRWISE.
Resources	Processor Time	00:00:00.03
	Elapsed Time	00:00:00.02
	Number of Cases Allowed	349525 cases[a]

a. Based on availability of workspace memory

Correlations

			Survival time in days	Partial residual for Year when accepted surgery
Spearman's rho	Survival time in days	Correlation Coefficient	1.000	.141
		Sig. (2-tailed)	.	.084
		N	206	150
	Partial residual for Year when accepted surgery	Correlation Coefficient	.141	1.000
		Sig. (2-tailed)	.084	.
		N	150	150
	Partial residual for Patient's age	Correlation Coefficient	.188*	-.040
		Sig. (2-tailed)	.021	.631
		N	150	150
	Partial residual for Received surgical care	Correlation Coefficient	-.627**	.122
		Sig. (2-tailed)	.000	.137
		N	150	150
	Partial residual for Received heart transplant	Correlation Coefficient	.110	.117
		Sig. (2-tailed)	.180	.155
		N	150	150
	Partial residual for Waiting time before surgery	Correlation Coefficient	.168*	.101
		Sig. (2-tailed)	.039	.219
		N	150	150

Correlations

			Partial residual for Patient's age	Partial residual for Received surgical care
Spearman's rho	Survival time in days	Correlation Coefficient	.188*	-.627**
		Sig. (2-tailed)	.021	.000
		N	150	150
	Partial residual for Year when accepted surgery	Correlation Coefficient	-.040	.122
		Sig. (2-tailed)	.631	.137
		N	150	150
	Partial residual for Patient's age	Correlation Coefficient	1.000	-.087
		Sig. (2-tailed)	.	.290
		N	150	150
	Partial residual for Received surgical care	Correlation Coefficient	-.087	1.000
		Sig. (2-tailed)	.290	.
		N	150	150
	Partial residual for Received heart transplant	Correlation Coefficient	.282**	-.115
		Sig. (2-tailed)	.000	.161
		N	150	150
	Partial residual for Waiting time before surgery	Correlation Coefficient	.316**	-.068
		Sig. (2-tailed)	.000	.406
		N	150	150

Correlations

			Partial residual for Received heart transplant	Partial residual for Waiting time before surgery
Spearman's rho	Survival time in days	Correlation Coefficient	.110	.168*
		Sig. (2-tailed)	.180	.039
		N	150	150
	Partial residual for Year when accepted surgery	Correlation Coefficient	.117	.101
		Sig. (2-tailed)	.155	.219
		N	150	150
	Partial residual for Patient's age	Correlation Coefficient	.282**	.316**
		Sig. (2-tailed)	.000	.000
		N	150	150
	Partial residual for Received surgical care	Correlation Coefficient	-.115	-.068
		Sig. (2-tailed)	.161	.406
		N	150	150
	Partial residual for Received heart transplant	Correlation Coefficient	1.000	.474**
		Sig. (2-tailed)	.	.000
		N	150	150
	Partial residual for Waiting time before surgery	Correlation Coefficient	.474**	1.000
		Sig. (2-tailed)	.000	.
		N	150	150

*. Correlation is significant at the 0.05 level (2-tailed).

**. Correlation is significant at the 0.01 level (2-tailed).

```
1325  0
1326  0  *Partial residuals were correlated with survival time (Pearson and Spearman).
1327  0      *Year of surgery showed no evidence of a time-varying effect.
1328  0      *Age was not significant on both tests (r  0.18).
1329  0      *Surgical care was significant but with opposite signs, so the result is unr
eliable for a binary covariate.
1330  0     * Heart transplant and waiting time gave discordant results.
1331  0      *Because the correlation approach is only suggestive, proportional hazards f
or age, surgical care, heart transplant and
             waiting time will be assessed formally with time-dependent covariates.
1332  0
1333  0  DATASET ACTIVATE DataSet1.
1334  0  TIME PROGRAM.
1335  0  COMPUTE T_COV_ = T_ * Year_surgery.
```

```
1336  0  COXREG   Survtime
1337  0    /STATUS=Status(1)
1338  0    /CONTRAST (Heart_transplant)=Indicator(1)
1339  0    /CONTRAST (Surg_care)=Indicator(1)
1340  0    /METHOD=ENTER Year_surgery Age Surg_care Heart_transplant Wait_time T_COV_
1341  0    /CRITERIA=PIN(.05) POUT(.10) ITERATE(20).
```

Cox Regression

Notes

Output Created		03-NOV-2026 15:35:56
Comments		
Input	Active Dataset	DataSet1
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	206
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
Syntax		TIME PROGRAM. COMPUTE T_COV_ = T_ * Year_surgery. COXREG Survtime /STATUS=Status(1) /CONTRAST (Heart_transplant) =Indicator(1) /CONTRAST (Surg_care) =Indicator(1) /METHOD=ENTER Year_surgery Age Surg_care Heart_transplant Wait_time T_COV_ /CRITERIA=PIN(.05) POUT(.10) ITERATE(20).
Resources	Processor Time	00:00:00.08
	Elapsed Time	00:00:00.08

Case Processing Summary

		N	Percent
Cases available in analysis	Event[a]	150	72.8%
	Censored	56	27.2%
	Total	206	100.0%
Cases dropped	Cases with missing values	0	0.0%
	Cases with negative time	0	0.0%
	Censored cases before the earliest event in a stratum	0	0.0%
	Total	0	0.0%
Total		206	100.0%

a. Dependent Variable: Survival time in days

Categorical Variable Codings[a,c]

		Frequency	(1)
Received surgical care[b]	0=No	182	0
	1=Yes	24	1
Received heart transplant[b]	0=No	68	0
	1=Yes	138	1

a. Category variable: Received surgical care (Surg_care)

b. Indicator Parameter Coding

c. Category variable: Received heart transplant (Heart_transplant)

Block 0: Beginning Block

Omnibus Tests of Model Coefficients

-2 Log Likelihood
1401.205

Block 1: Method = Enter

Omnibus Tests of Model Coefficients[a]

-2 Log Likelihood	Overall (score)			Change From Previous Step			Change ...
	Chi-square	df	Sig.	Chi-square	df	Sig.	Chi-square
1290.144	117.820	6	.000	111.061	6	.000	111.061

Omnibus Tests of Model Coefficients[a]

Change From Previous ..	
df	Sig.
6	.000

a. Beginning Block Number 1. Method = Enter

Variables in the Equation

	B	SE	Wald	df	Sig.	Exp(B)
Year when accepted surgery	-.150	.056	7.213	1	.007	.861
Patient's age	.056	.010	30.444	1	.000	1.058
Received surgical care	-.858	.342	6.298	1	.012	.424
Received heart transplant	-1.344	.222	36.716	1	.000	.261
Waiting time before surgery	-.007	.004	4.476	1	.034	.993
T_COV_	.001	.000	6.385	1	.012	1.001

Covariate Means

	Mean
Year when accepted surgery	70.793
Patient's age	43.961
Received surgical care	.176
Received heart transplant	.825
Waiting time before surgery	35.831
T_COV_	6170.013

```
1342  0
1343  0  *T_ × Year, p = .012 ; year possibly violate sph
1344  0  *Comment: T_COV_ B = .001 (positive), Wald 6.385. Year's protective effect (HR .
861 at time 0)
1345  0  *fades as follow-up lengthens. On 206 rows p = .012; on 103 unique patients it w
ould be about .07.
1346  0  *Verdict: possible violation, fragile.
```

```
1347  0
1348  0
1349  0
1350  0  TIME PROGRAM.
1351  0  COMPUTE T_COV_ = T_ * Age.
1352  0  COXREG   Survtime
1353  0    /STATUS=Status(1)
1354  0    /CONTRAST (Heart_transplant)=Indicator(1)
1355  0    /CONTRAST (Surg_care)=Indicator(1)
1356  0    /METHOD=ENTER Year_surgery Age Surg_care Heart_transplant Wait_time T_COV_
1357  0    /CRITERIA=PIN(.05) POUT(.10) ITERATE(20).
```

Cox Regression

Notes

Output Created		03-NOV-2026 15:35:56
Comments		
Input	Active Dataset	DataSet1
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	206
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
Syntax		TIME PROGRAM. COMPUTE T_COV_ = T_ * Age. COXREG Survtime /STATUS=Status(1) /CONTRAST (Heart_transplant) =Indicator(1) /CONTRAST (Surg_care) =Indicator(1) /METHOD=ENTER Year_surgery Age Surg_care Heart_transplant Wait_time T_COV_ /CRITERIA=PIN(.05) POUT(.10) ITERATE(20).
Resources	Processor Time	00:00:00.05
	Elapsed Time	00:00:00.09

Case Processing Summary

		N	Percent
Cases available in analysis	Event[a]	150	72.8%
	Censored	56	27.2%
	Total	206	100.0%
Cases dropped	Cases with missing values	0	0.0%
	Cases with negative time	0	0.0%
	Censored cases before the earliest event in a stratum	0	0.0%
	Total	0	0.0%
Total		206	100.0%

a. Dependent Variable: Survival time in days

Categorical Variable Codings[a,c]

		Frequency	(1)
Received surgical care[b]	0=No	182	0
	1=Yes	24	1
Received heart transplant[b]	0=No	68	0
	1=Yes	138	1

a. Category variable: Received surgical care (Surg_care)

b. Indicator Parameter Coding

c. Category variable: Received heart transplant (Heart_transplant)

Block 0: Beginning Block

Omnibus Tests of Model Coefficients

-2 Log Likelihood
1401.205

Block 1: Method = Enter

Omnibus Tests of Model Coefficients[a]

-2 Log Likelihood	Overall (score)			Change From Previous Step			Change ...
	Chi-square	df	Sig.	Chi-square	df	Sig.	Chi-square
1290.272	120.365	6	.000	110.933	6	.000	110.933

Omnibus Tests of Model Coefficients[a]

Change From Previous ..	
df	Sig.
6	.000

a. Beginning Block Number 1. Method = Enter

Variables in the Equation

	B	SE	Wald	df	Sig.	Exp(B)
Year when accepted surgery	-.073	.047	2.441	1	.118	.930
Patient's age	.043	.011	14.707	1	.000	1.044
Received surgical care	-.635	.318	3.988	1	.046	.530
Received heart transplant	-1.424	.226	39.537	1	.000	.241
Waiting time before surgery	-.008	.004	4.426	1	.035	.992
T_COV_	.000	.000	5.193	1	.023	1.000

Covariate Means

	Mean
Year when accepted surgery	70.793
Patient's age	43.961
Received surgical care	.176
Received heart transplant	.825
Waiting time before surgery	35.831
T_COV_	3788.201

```
1358  0
1359  0  *T_ × Age .023 age possibley violates
1360  0  *Comment: T_COV_ Wald 5.193. B shows as .000 because age is in years and time in
 days,
1361  0  *so direction not read. On 206 rows p = .023; on 103 unique patients about .11.
1362  0  *Verdict: possible violation, fragile.
1363  0
```

```
1364  0
1365  0
1366  0  TIME PROGRAM.
1367  0  COMPUTE T_COV_ = T_ * Surg_care.
1368  0  COXREG   Survtime
1369  0    /STATUS=Status(1)
1370  0    /CONTRAST (Heart_transplant)=Indicator(1)
1371  0    /CONTRAST (Surg_care)=Indicator(1)
1372  0    /METHOD=ENTER Year_surgery Age Surg_care Heart_transplant Wait_time T_COV_
1373  0    /CRITERIA=PIN(.05) POUT(.10) ITERATE(20).
```

Cox Regression

Notes

Output Created		03-NOV-2026 15:35:56
Comments		
Input	Active Dataset	DataSet1
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	206
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
Syntax		TIME PROGRAM. COMPUTE T_COV_ = T_ * Surg_care. COXREG Survtime /STATUS=Status(1) /CONTRAST (Heart_transplant) =Indicator(1) /CONTRAST (Surg_care) =Indicator(1) /METHOD=ENTER Year_surgery Age Surg_care Heart_transplant Wait_time T_COV_ /CRITERIA=PIN(.05) POUT(.10) ITERATE(20).
Resources	Processor Time	00:00:00.05
	Elapsed Time	00:00:00.07

Case Processing Summary

		N	Percent
Cases available in analysis	Event[a]	150	72.8%
	Censored	56	27.2%
	Total	206	100.0%
Cases dropped	Cases with missing values	0	0.0%
	Cases with negative time	0	0.0%
	Censored cases before the earliest event in a stratum	0	0.0%
	Total	0	0.0%
Total		206	100.0%

a. Dependent Variable: Survival time in days

Categorical Variable Codings[a,c]

		Frequency	(1)
Received surgical care[b]	0=No	182	0
	1=Yes	24	1
Received heart transplant[b]	0=No	68	0
	1=Yes	138	1

a. Category variable: Received surgical care (Surg_care)

b. Indicator Parameter Coding

c. Category variable: Received heart transplant (Heart_transplant)

Block 0: Beginning Block

Omnibus Tests of Model Coefficients

-2 Log Likelihood
1401.205

Block 1: Method = Enter

Omnibus Tests of Model Coefficients[a]

-2 Log Likelihood	Overall (score)			Change From Previous Step			Change ...
	Chi-square	df	Sig.	Chi-square	df	Sig.	Chi-square
1290.183	115.708	6	.000	111.021	6	.000	111.021

Omnibus Tests of Model Coefficients[a]

Change From Previous ..	
df	Sig.
6	.000

a. Beginning Block Number 1. Method = Enter

Variables in the Equation

	B	SE	Wald	df	Sig.	Exp(B)
Year when accepted surgery	-.080	.047	2.968	1	.085	.923
Patient's age	.055	.010	29.269	1	.000	1.056
Received surgical care	-1.359	.491	7.656	1	.006	.257
Received heart transplant	-1.357	.223	36.934	1	.000	.257
Waiting time before surgery	-.007	.004	4.349	1	.037	.993
T_COV_	.002	.001	6.424	1	.011	1.002

Covariate Means

	Mean
Year when accepted surgery	70.793
Patient's age	43.961
Received surgical care	.176
Received heart transplant	.825
Waiting time before surgery	35.831
T_COV_	21.739

```
1374  0
1375  0  *Comment: T_COV_ B = .002 (positive), Wald 6.424, p = .011. The benefit of surgi
cal care
1376  0  *(HR .257 at time 0) fades as follow-up lengthens. About .07 on 103 unique patie
nts, and the
1377  0  *trend rests on only 12 patients with surgical care. Verdict: possible violation
, least reliable.
```

```
1378  0
1379  0
1380  0  TIME PROGRAM.
1381  0  COMPUTE T_COV_ = T_ * Heart_transplant.
1382  0  COXREG   Survtime
1383  0    /STATUS=Status(1)
1384  0    /CONTRAST (Heart_transplant)=Indicator(1)
1385  0    /CONTRAST (Surg_care)=Indicator(1)
1386  0    /METHOD=ENTER Year_surgery Age Surg_care Heart_transplant Wait_time T_COV_
1387  0    /CRITERIA=PIN(.05) POUT(.10) ITERATE(20).
```

Cox Regression

Notes

Output Created		03-NOV-2026 15:35:56
Comments		
Input	Active Dataset	DataSet1
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	206
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
Syntax		TIME PROGRAM. COMPUTE T_COV_ = T_ * Heart_transplant. COXREG Survtime /STATUS=Status(1) /CONTRAST (Heart_transplant) =Indicator(1) /CONTRAST (Surg_care) =Indicator(1) /METHOD=ENTER Year_surgery Age Surg_care Heart_transplant Wait_time T_COV_ /CRITERIA=PIN(.05) POUT(.10) ITERATE(20).
Resources	Processor Time	00:00:00.06
	Elapsed Time	00:00:00.06

Case Processing Summary

		N	Percent
Cases available in analysis	Event[a]	150	72.8%
	Censored	56	27.2%
	Total	206	100.0%
Cases dropped	Cases with missing values	0	0.0%
	Cases with negative time	0	0.0%
	Censored cases before the earliest event in a stratum	0	0.0%
	Total	0	0.0%
Total		206	100.0%

a. Dependent Variable: Survival time in days

Categorical Variable Codings[a,c]

		Frequency	(1)
Received surgical care[b]	0=No	182	0
	1=Yes	24	1
Received heart transplant[b]	0=No	68	0
	1=Yes	138	1

a. Category variable: Received surgical care (Surg_care)

b. Indicator Parameter Coding

c. Category variable: Received heart transplant (Heart_transplant)

Block 0: Beginning Block

Omnibus Tests of Model Coefficients

-2 Log Likelihood
1401.205

Block 1: Method = Enter

Omnibus Tests of Model Coefficients[a]

-2 Log Likelihood	Overall (score)			Change From Previous Step			Change ...
	Chi-square	df	Sig.	Chi-square	df	Sig.	Chi-square
1286.228	123.860	6	.000	114.977	6	.000	114.977

Omnibus Tests of Model Coefficients[a]

Change From Previous ..	
df	Sig.
6	.000

a. Beginning Block Number 1. Method = Enter

Variables in the Equation

	B	SE	Wald	df	Sig.	Exp(B)
Year when accepted surgery	-.068	.047	2.087	1	.149	.935
Patient's age	.051	.010	25.628	1	.000	1.052
Received surgical care	-.696	.319	4.773	1	.029	.499
Received heart transplant	-1.759	.271	42.085	1	.000	.172
Waiting time before surgery	-.008	.004	4.660	1	.031	.992
T_COV_	.004	.002	4.251	1	.039	1.004

Covariate Means

	Mean
Year when accepted surgery	70.793
Patient's age	43.961
Received surgical care	.176
Received heart transplant	.825
Waiting time before surgery	35.831
T_COV_	77.786

```
1388  0
1389  0  *Comment: T_COV_ B = .004 (positive), Wald 4.251, p = .039. The benefit of trans
plant
1390  0  *(HR .172 at time 0) fades as follow-up lengthens. About .15 on 103 unique patie
nts.
1391  0  *This tests the fixed 0/1 coding, so immortal-time bias could produce the same p
attern.
```

```
1392  0  *Verdict: possible violation, interpret with caution.
1393  0
1394  0  TIME PROGRAM.
1395  0  COMPUTE T_COV_ = T_ * Wait_time.
1396  0  COXREG   Survtime
1397  0    /STATUS=Status(1)
1398  0    /CONTRAST (Heart_transplant)=Indicator(1)
1399  0    /CONTRAST (Surg_care)=Indicator(1)
1400  0    /METHOD=ENTER Year_surgery Age Surg_care Heart_transplant Wait_time T_COV_
1401  0    /CRITERIA=PIN(.05) POUT(.10) ITERATE(20).
```

Cox Regression

Notes

Output Created		03-NOV-2026 15:35:56
Comments		
Input	Active Dataset	DataSet1
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	206
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
Syntax		TIME PROGRAM. COMPUTE T_COV_ = T_ * Wait_time. COXREG Survtime /STATUS=Status(1) /CONTRAST (Heart_transplant) =Indicator(1) /CONTRAST (Surg_care) =Indicator(1) /METHOD=ENTER Year_surgery Age Surg_care Heart_transplant Wait_time T_COV_ /CRITERIA=PIN(.05) POUT(.10) ITERATE(20).
Resources	Processor Time	00:00:00.08
	Elapsed Time	00:00:00.05

Case Processing Summary

		N	Percent
Cases available in analysis	Event[a]	150	72.8%
	Censored	56	27.2%
	Total	206	100.0%
Cases dropped	Cases with missing values	0	0.0%
	Cases with negative time	0	0.0%
	Censored cases before the earliest event in a stratum	0	0.0%
	Total	0	0.0%
Total		206	100.0%

a. Dependent Variable: Survival time in days

Categorical Variable Codings[a,c]

		Frequency	(1)
Received surgical care[b]	0=No	182	0
	1=Yes	24	1
Received heart transplant[b]	0=No	68	0
	1=Yes	138	1

a. Category variable: Received surgical care (Surg_care)

b. Indicator Parameter Coding

c. Category variable: Received heart transplant (Heart_transplant)

Block 0: Beginning Block

Omnibus Tests of Model Coefficients

-2 Log Likelihood
1401.205

Block 1: Method = Enter

Omnibus Tests of Model Coefficients[a]

-2 Log Likelihood	Overall (score)			Change From Previous Step			Change ...
	Chi-square	df	Sig.	Chi-square	df	Sig.	Chi-square
1290.469	115.239	6	.000	110.735	6	.000	110.735

Omnibus Tests of Model Coefficients[a]

Change From Previous ..	
df	Sig.
6	.000

a. Beginning Block Number 1. Method = Enter

Variables in the Equation

	B	SE	Wald	df	Sig.	Exp(B)
Year when accepted surgery	-.075	.047	2.573	1	.109	.928
Patient's age	.054	.010	27.940	1	.000	1.055
Received surgical care	-.689	.321	4.607	1	.032	.502
Received heart transplant	-1.317	.227	33.531	1	.000	.268
Waiting time before surgery	-.014	.005	8.060	1	.005	.986
T_COV_	.000	.000	5.594	1	.018	1.000

Covariate Means

	Mean
Year when accepted surgery	70.793
Patient's age	43.961
Received surgical care	.176
Received heart transplant	.825
Waiting time before surgery	35.831
T_COV_	3399.166

```
1402  0
1403  0  *Comment: T_COV_ Wald 5.594, p = .018; B shows as .000 (days by days), so direct
ion not read.
1404  0  *About .09 on 103 unique patients. Waiting time is 0 for everyone without a tran
splant,
1405  0  *so its trend overlaps with transplant. Verdict: possible violation, fragile.
1406  0
```

```
1407  0  TIME PROGRAM.
1408  0  COMPUTE T_COV_1 = T_ * Wait_time.
1409  0  COMPUTE T_COV_2 = T_ * Age.
1410  0  COMPUTE T_COV_3 = T_ * Surg_care.
1411  0  COMPUTE T_COV_4 = T_ * Heart_transplant.
1412  0  COMPUTE T_COV_5 = T_ * Year_surgery.
1413  0  COXREG Survtime
1414  0    /STATUS=Status(1)
1415  0    /CONTRAST (Heart_transplant)=Indicator(1)
1416  0    /CONTRAST (Surg_care)=Indicator(1)
1417  0    /METHOD=ENTER Year_surgery Age Surg_care Heart_transplant Wait_time T_COV_1 T_
COV_2 T_COV_3 T_COV_4 T_COV_5
1418  0    /CRITERIA=PIN(.05) POUT(.10) ITERATE(20).
```

Cox Regression

Notes

Output Created		03-NOV-2026 15:35:56
Comments		
Input	Active Dataset	DataSet1
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	206
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.

Notes

Syntax		TIME PROGRAM. COMPUTE T_COV_1 = T_ * Wait_time. COMPUTE T_COV_2 = T_ * Age. COMPUTE T_COV_3 = T_ * Surg_care. COMPUTE T_COV_4 = T_ * Heart_transplant. COMPUTE T_COV_5 = T_ * Year_surgery. COXREG Survtime /STATUS=Status(1) /CONTRAST (Heart_transplant) =Indicator(1) /CONTRAST (Surg_care) =Indicator(1) /METHOD=ENTER Year_surgery Age Surg_care Heart_transplant Wait_time T_COV_1 T_COV_2 T_COV_3 T_COV_4 T_COV_5 /CRITERIA=PIN(.05) POUT(.10) ITERATE(20).
Resources	Processor Time	00:00:00.11
	Elapsed Time	00:00:00.10

Case Processing Summary

		N	Percent
Cases available in analysis	Event[a]	150	72.8%
	Censored	56	27.2%
	Total	206	100.0%
Cases dropped	Cases with missing values	0	0.0%
	Cases with negative time	0	0.0%
	Censored cases before the earliest event in a stratum	0	0.0%
	Total	0	0.0%
Total		206	100.0%

a. Dependent Variable: Survival time in days

Categorical Variable Codings[a,c]

		Frequency	(1)
Received surgical care[b]	0=No	182	0
	1=Yes	24	1
Received heart transplant[b]	0=No	68	0
	1=Yes	138	1

a. Category variable: Received surgical care (Surg_care)

b. Indicator Parameter Coding

c. Category variable: Received heart transplant (Heart_transplant)

Block 0: Beginning Block

Omnibus Tests of Model Coefficients

-2 Log Likelihood
1401.205

Block 1: Method = Enter

Omnibus Tests of Model Coefficients[a]

-2 Log Likelihood	Overall (score)			Change From Previous Step			Change ...
	Chi-square	df	Sig.	Chi-square	df	Sig.	Chi-square
1276.328	130.862	10	.000	124.877	10	.000	124.877

Omnibus Tests of Model Coefficients[a]

Change From Previous ..	
df	Sig.
10	.000

a. Beginning Block Number 1. Method = Enter

Variables in the Equation

	B	SE	Wald	df	Sig.	Exp(B)
Year when accepted surgery	-.139	.058	5.687	1	.017	.870
Patient's age	.042	.011	13.342	1	.000	1.043
Received surgical care	-1.171	.498	5.526	1	.019	.310
Received heart transplant	-1.589	.292	29.648	1	.000	.204
Waiting time before surgery	-.010	.005	4.220	1	.040	.990
T_COV_1	.000	.000	.595	1	.440	1.000
T_COV_2	.000	.000	2.560	1	.110	1.000
T_COV_3	.000	.001	.199	1	.656	1.000
T_COV_4	.004	.002	2.052	1	.152	1.004
T_COV_5	.001	.000	4.316	1	.038	1.001

Covariate Means

	Mean
Year when accepted surgery	70.793
Patient's age	43.961
Received surgical care	.176
Received heart transplant	.825
Waiting time before surgery	35.831
T_COV_1	3399.166
T_COV_2	3788.201
T_COV_3	21.739
T_COV_4	77.786
T_COV_5	6170.013

```
1419  0
1420  0  *Comment: all five time-dependent terms together (1 wait time, 2 age, 3 surgical
 care,
1421  0  *4 transplant, 5 year); -2LL = 1276.328. P-values: T_COV_1 .440, T_COV_2 .110, T
_COV_3 .656,
1422  0  *T_COV_4 .152, T_COV_5 .038. Only year of surgery stays significant; the other t
rends overlap
1423  0  *and lose significance. Verdict: drop the least significant term, T_COV_3 (surgi
cal care).
1424  0
1425  0  *drop least signficant time varying covariate surg care
```

```
1426  0
1427  0  TIME PROGRAM.
1428  0  COMPUTE T_COV_1 = T_ * Wait_time.
1429  0  COMPUTE T_COV_2 = T_ * Age.
1430  0  COMPUTE T_COV_4 = T_ * Heart_transplant.
1431  0  COMPUTE T_COV_5 = T_ * Year_surgery.
1432  0  COXREG Survtime
1433  0    /STATUS=Status(1)
1434  0    /CONTRAST (Heart_transplant)=Indicator(1)
1435  0    /CONTRAST (Surg_care)=Indicator(1)
1436  0    /METHOD=ENTER Year_surgery Age Surg_care Heart_transplant Wait_time T_COV_1 T_
COV_2 T_COV_4 T_COV_5
1437  0    /CRITERIA=PIN(.05) POUT(.10) ITERATE(20).
```

Cox Regression

Notes

Output Created		03-NOV-2026 15:35:56
Comments		
Input	Active Dataset	DataSet1
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	206
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.

Notes

Syntax		TIME PROGRAM. COMPUTE T_COV_1 = T_ * Wait_time. COMPUTE T_COV_2 = T_ * Age. COMPUTE T_COV_4 = T_ * Heart_transplant. COMPUTE T_COV_5 = T_ * Year_surgery. COXREG Survtime /STATUS=Status(1) /CONTRAST (Heart_transplant) =Indicator(1) /CONTRAST (Surg_care) =Indicator(1) /METHOD=ENTER Year_surgery Age Surg_care Heart_transplant Wait_time T_COV_1 T_COV_2 T_COV_4 T_COV_5 /CRITERIA=PIN(.05) POUT(.10) ITERATE(20).
Resources	Processor Time	00:00:00.08
	Elapsed Time	00:00:00.09

Case Processing Summary

		N	Percent
Cases available in analysis	Event[a]	150	72.8%
	Censored	56	27.2%
	Total	206	100.0%
Cases dropped	Cases with missing values	0	0.0%
	Cases with negative time	0	0.0%
	Censored cases before the earliest event in a stratum	0	0.0%
	Total	0	0.0%
Total		206	100.0%

a. Dependent Variable: Survival time in days

Categorical Variable Codings[a,c]

		Frequency	(1)
Received surgical care[b]	0=No	182	0
	1=Yes	24	1
Received heart transplant[b]	0=No	68	0
	1=Yes	138	1

a. Category variable: Received surgical care (Surg_care)

b. Indicator Parameter Coding

c. Category variable: Received heart transplant (Heart_transplant)

Block 0: Beginning Block

Omnibus Tests of Model Coefficients

-2 Log Likelihood
1401.205

Block 1: Method = Enter

Omnibus Tests of Model Coefficients[a]

-2 Log Likelihood	Overall (score)			Change From Previous Step			Change ...
	Chi-square	df	Sig.	Chi-square	df	Sig.	Chi-square
1276.528	130.483	9	.000	124.677	9	.000	124.677

Omnibus Tests of Model Coefficients[a]

Change From Previous ..	
df	Sig.
9	.000

a. Beginning Block Number 1. Method = Enter

Variables in the Equation

	B	SE	Wald	df	Sig.	Exp(B)
Year when accepted surgery	-.147	.056	6.995	1	.008	.863
Patient's age	.041	.011	13.272	1	.000	1.042
Received surgical care	-1.022	.355	8.274	1	.004	.360
Received heart transplant	-1.594	.292	29.839	1	.000	.203
Waiting time before surgery	-.010	.005	4.252	1	.039	.990
T_COV_1	.000	.000	.596	1	.440	1.000
T_COV_2	.000	.000	2.889	1	.089	1.000
T_COV_4	.004	.002	2.083	1	.149	1.004
T_COV_5	.001	.000	7.967	1	.005	1.001

Covariate Means

	Mean
Year when accepted surgery	70.793
Patient's age	43.961
Received surgical care	.176
Received heart transplant	.825
Waiting time before surgery	35.831
T_COV_1	3399.166
T_COV_2	3788.201
T_COV_4	77.786
T_COV_5	6170.013

```
1438  0
1439  0  *Comment: -2LL = 1276.528, a change of 0.2 on 1 df from the five-term model (p a
bout .65),
1440  0  *so dropping T_COV_3 cost nothing. P-values: T_COV_1 .440, T_COV_2 .089, T_COV_4
 .149,
1441  0  *T_COV_5 .005. Verdict: drop T_COV_1 (waiting time) next.
1442  0
1443  0  *drop wait time
1444  0
1445  0  TIME PROGRAM.
1446  0  COMPUTE T_COV_2 = T_ * Age.
1447  0  COMPUTE T_COV_4 = T_ * Heart_transplant.
1448  0  COMPUTE T_COV_5 = T_ * Year_surgery.
1449  0  COXREG Survtime
```

```
1450  0    /STATUS=Status(1)
1451  0    /CONTRAST (Heart_transplant)=Indicator(1)
1452  0    /CONTRAST (Surg_care)=Indicator(1)
1453  0    /METHOD=ENTER Year_surgery Age Surg_care Heart_transplant Wait_time T_COV_2 T_
COV_4 T_COV_5
1454  0    /CRITERIA=PIN(.05) POUT(.10) ITERATE(20).
```

Cox Regression

Notes

Output Created		03-NOV-2026 15:35:56
Comments		
Input	Active Dataset	DataSet1
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	206
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
Syntax		TIME PROGRAM. COMPUTE T_COV_2 = T_ * Age. COMPUTE T_COV_4 = T_ * Heart_transplant. COMPUTE T_COV_5 = T_ * Year_surgery. COXREG Survtime /STATUS=Status(1) /CONTRAST (Heart_transplant) =Indicator(1) /CONTRAST (Surg_care) =Indicator(1) /METHOD=ENTER Year_surgery Age Surg_care Heart_transplant Wait_time T_COV_2 T_COV_4 T_COV_5 /CRITERIA=PIN(.05) POUT(.10) ITERATE(20).
Resources	Processor Time	00:00:00.05
	Elapsed Time	00:00:00.06

Case Processing Summary

		N	Percent
Cases available in analysis	Event[a]	150	72.8%
	Censored	56	27.2%
	Total	206	100.0%
Cases dropped	Cases with missing values	0	0.0%
	Cases with negative time	0	0.0%
	Censored cases before the earliest event in a stratum	0	0.0%
	Total	0	0.0%
Total		206	100.0%

a. Dependent Variable: Survival time in days

Categorical Variable Codings[a,c]

		Frequency	(1)
Received surgical care[b]	0=No	182	0
	1=Yes	24	1
Received heart transplant[b]	0=No	68	0
	1=Yes	138	1

a. Category variable: Received surgical care (Surg_care)

b. Indicator Parameter Coding

c. Category variable: Received heart transplant (Heart_transplant)

Block 0: Beginning Block

Omnibus Tests of Model Coefficients

-2 Log Likelihood
1401.205

Block 1: Method = Enter

Omnibus Tests of Model Coefficients[a]

-2 Log Likelihood	Overall (score)			Change From Previous Step			Change ...
	Chi-square	df	Sig.	Chi-square	df	Sig.	Chi-square
1277.130	130.197	8	.000	124.075	8	.000	124.075

Omnibus Tests of Model Coefficients[a]

Change From Previous ..	
df	Sig.
8	.000

a. Beginning Block Number 1. Method = Enter

Variables in the Equation

	B	SE	Wald	df	Sig.	Exp(B)
Year when accepted surgery	-.148	.055	7.151	1	.007	.862
Patient's age	.042	.011	13.456	1	.000	1.042
Received surgical care	-1.029	.358	8.278	1	.004	.357
Received heart transplant	-1.660	.279	35.268	1	.000	.190
Waiting time before surgery	-.008	.004	4.706	1	.030	.992
T_COV_2	.000	.000	2.953	1	.086	1.000
T_COV_4	.004	.002	2.618	1	.106	1.004
T_COV_5	.001	.000	8.312	1	.004	1.001

Covariate Means

	Mean
Year when accepted surgery	70.793
Patient's age	43.961
Received surgical care	.176
Received heart transplant	.825
Waiting time before surgery	35.831
T_COV_2	3788.201
T_COV_4	77.786
T_COV_5	6170.013

```
1455  0
```

```
1456  0  *Comment: P-values: T_COV_2 .086, T_COV_4 .106, T_COV_5 .004. Verdict: drop T_CO
V_4 (transplant) next.
1457  0  *This removes the trend term only; the fixed transplant coding still needs to be
 flagged.
1458  0
1459  0  *drop transplant
1460  0
1461  0  TIME PROGRAM.
1462  0  COMPUTE T_COV_2 = T_ * Age.
1463  0  COMPUTE T_COV_5 = T_ * Year_surgery.
1464  0  COXREG Survtime
1465  0    /STATUS=Status(1)
1466  0    /CONTRAST (Heart_transplant)=Indicator(1)
1467  0    /CONTRAST (Surg_care)=Indicator(1)
1468  0    /METHOD=ENTER Year_surgery Age Surg_care Heart_transplant Wait_time T_COV_2 T_
COV_5
1469  0    /CRITERIA=PIN(.05) POUT(.10) ITERATE(20).
```

Cox Regression

Notes

Output Created		03-NOV-2026 15:35:56
Comments		
Input	Active Dataset	DataSet1
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	206
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.

Notes

Syntax		TIME PROGRAM. COMPUTE T_COV_2 = T_ * Age. COMPUTE T_COV_5 = T_ * Year_surgery. COXREG Survtime /STATUS=Status(1) /CONTRAST (Heart_transplant) =Indicator(1) /CONTRAST (Surg_care) =Indicator(1) /METHOD=ENTER Year_surgery Age Surg_care Heart_transplant Wait_time T_COV_2 T_COV_5 /CRITERIA=PIN(.05) POUT(.10) ITERATE(20).
Resources	Processor Time	00:00:00.06
	Elapsed Time	00:00:00.06

Case Processing Summary

		N	Percent
Cases available in analysis	Event[a]	150	72.8%
	Censored	56	27.2%
	Total	206	100.0%
Cases dropped	Cases with missing values	0	0.0%
	Cases with negative time	0	0.0%
	Censored cases before the earliest event in a stratum	0	0.0%
	Total	0	0.0%
Total		206	100.0%

a. Dependent Variable: Survival time in days

Categorical Variable Codings[a,c]

		Frequency	(1)
Received surgical care[b]	0=No	182	0
	1=Yes	24	1
Received heart transplant[b]	0=No	68	0
	1=Yes	138	1

a. Category variable: Received surgical care (Surg_care)

b. Indicator Parameter Coding

c. Category variable: Received heart transplant (Heart_transplant)

Block 0: Beginning Block

Omnibus Tests of Model Coefficients

-2 Log Likelihood
1401.205

Block 1: Method = Enter

Omnibus Tests of Model Coefficients[a]

-2 Log Likelihood	Overall (score)			Change From Previous Step			Change ...
	Chi-square	df	Sig.	Chi-square	df	Sig.	Chi-square
1280.797	127.571	7	.000	120.408	7	.000	120.408

Omnibus Tests of Model Coefficients[a]

Change From Previous ..	
df	Sig.
7	.000

a. Beginning Block Number 1. Method = Enter

Variables in the Equation

	B	SE	Wald	df	Sig.	Exp(B)
Year when accepted surgery	-.161	.055	8.487	1	.004	.851
Patient's age	.039	.011	12.401	1	.000	1.040
Received surgical care	-.989	.358	7.633	1	.006	.372
Received heart transplant	-1.377	.224	37.648	1	.000	.252
Waiting time before surgery	-.007	.003	4.312	1	.038	.993
T_COV_2	.000	.000	7.265	1	.007	1.000
T_COV_5	.001	.000	9.640	1	.002	1.001

Covariate Means

	Mean
Year when accepted surgery	70.793
Patient's age	43.961
Received surgical care	.176
Received heart transplant	.825
Waiting time before surgery	35.831
T_COV_2	3788.201
T_COV_5	6170.013

```
1470  0
1471  0  *Comment: -2LL = 1280.797 (7 df); against the nine-term model the change is 4.27
 on 2 df (p about .12),
1472  0  *so removing both terms did not worsen the fit. Both remaining terms are signifi
cant:
1473  0  *T_COV_2 (age) Wald 7.265, p = .007; T_COV_5 (year) Wald 9.640, p = .002.
1474  0  *Verdict: proportional hazards violated for age and year of surgery; keep both t
erms in all later models.
1475  0  *Main effects are now effects at time 0. P-values are too small because of the d
uplicated rows
1476  0  *(about .06 for age and .03 for year on 103 unique patients).
1477  0
1478  0  *CHECKING FUNCTIONAL TERMS
1479  0      *Check distributions first and do categorisation
1480  0
1481  0  FREQUENCIES VARIABLES=Year_surgery Age Wait_time
1482  0    /FORMAT=NOTABLE
1483  0    /STATISTICS=MINIMUM MAXIMUM MEAN MEDIAN
1484  0    /ORDER=ANALYSIS.
```

Frequencies

Notes

Output Created		03-NOV-2026 15:35:56
Comments		
Input	Active Dataset	DataSet1
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	206
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics are based on all cases with valid data.
Syntax		FREQUENCIES VARIABLES=Year_surgery Age Wait_time /FORMAT=NOTABLE /STATISTICS=MINIMUM MAXIMUM MEAN MEDIAN...
Resources	Processor Time	00:00:00.00
	Elapsed Time	00:00:00.01

Statistics

		Year when accepted surgery	Patient's age	Waiting time before surgery
N	Valid	206	206	206
	Missing	0	0	0
Mean		70.62	44.64	25.74
Median		71.00	47.00	10.00
Minimum		67	8	0
Maximum		74	64	310

```
1485  0
1486  0
1487  0  RECODE Year_surgery (Lowest thru 71=0) (72 thru Highest=1) INTO Year_SurgeryCat.
1488  0  RECODE Age (Lowest thru 47=0) (48 thru Highest=1) INTO AgeCat.
1489  0  RECODE Wait_time (Lowest thru 10=0) (10.01 thru Highest=1) INTO Wait_TimeCat.
1490  0  VARIABLE LABELS Year_SurgeryCat 'Year of surgery (median split)'
```

```
1491  0    /AgeCat 'Age (median split)'
1492  0    /Wait_TimeCat 'Waiting time (median split)'.
1493  0  VALUE LABELS Year_SurgeryCat 0 '71 or earlier' 1 '72 or later'
1494  0    /AgeCat 0 '47 or younger' 1 '48 or older'
1495  0    /Wait_TimeCat 0 '10 days or less' 1 'More than 10 days'.
1496  0  EXECUTE.
1497  0  FREQUENCIES VARIABLES=Year_SurgeryCat AgeCat Wait_TimeCat.
```

Frequencies

Notes

Output Created		03-NOV-2026 15:35:56
Comments		
Input	Active Dataset	DataSet1
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	206
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics are based on all cases with valid data.
Syntax		FREQUENCIES VARIABLES=Year_SurgeryCat AgeCat Wait_TimeCat.
Resources	Processor Time	00:00:00.00
	Elapsed Time	00:00:00.01

Statistics

		Year of surgery (median split)	Age (median split)	Waiting time (median split)
N	Valid	206	206	206
	Missing	0	0	0

Frequency Table

Year of surgery (median split)

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	71 or earlier	130	63.1	63.1	63.1
	72 or later	76	36.9	36.9	100.0
	Total	206	100.0	100.0	

Age (median split)

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	47 or younger	114	55.3	55.3	55.3
	48 or older	92	44.7	44.7	100.0
	Total	206	100.0	100.0	

Waiting time (median split)

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	10 days or less	106	51.5	51.5	51.5
	More than 10 days	100	48.5	48.5	100.0
	Total	206	100.0	100.0	

```
1498  0
1499  0  *model A (age categorised)
1500  0
1501  0  TIME PROGRAM.
1502  0  COMPUTE T_COV_2 = T_ * AgeCat.
1503  0  COMPUTE T_COV_5 = T_ * Year_surgery.
1504  0  COXREG Survtime
1505  0    /STATUS=Status(1)
1506  0    /CONTRAST (Heart_transplant)=Indicator(1)
1507  0    /CONTRAST (Surg_care)=Indicator(1)
1508  0    /CONTRAST (AgeCat)=Indicator(1)
1509  0    /METHOD=ENTER Year_surgery AgeCat Surg_care Heart_transplant Wait_time T_COV_2
 T_COV_5
1510  0    /CRITERIA=PIN(.05) POUT(.10) ITERATE(20).
```

Cox Regression

Notes

Output Created		03-NOV-2026 15:35:57
Comments		
Input	Active Dataset	DataSet1
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	206
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
Syntax		TIME PROGRAM. COMPUTE T_COV_2 = T_ * AgeCat. COMPUTE T_COV_5 = T_ * Year_surgery. COXREG Survtime /STATUS=Status(1) /CONTRAST (Heart_transplant) =Indicator(1) /CONTRAST (Surg_care) =Indicator(1) /CONTRAST (AgeCat) =Indicator(1) /METHOD=ENTER Year_surgery AgeCat Surg_care Heart_transplant Wait_time T_COV_2 T_COV_5 /CRITERIA=PIN(.05) POUT(.10) ITERATE(20).
Resources	Processor Time	00:00:00.06
	Elapsed Time	00:00:00.05

Case Processing Summary

		N	Percent
Cases available in analysis	Event[a]	150	72.8%
	Censored	56	27.2%
	Total	206	100.0%
Cases dropped	Cases with missing values	0	0.0%
	Cases with negative time	0	0.0%
	Censored cases before the earliest event in a stratum	0	0.0%
	Total	0	0.0%
Total		206	100.0%

a. Dependent Variable: Survival time in days

Categorical Variable Codings[a,c,d]

		Frequency	(1)
Received surgical care[b]	0=No	182	0
	1=Yes	24	1
Received heart transplant[b]	0=No	68	0
	1=Yes	138	1
Age (median split)[b]	.00=47 or younger	114	0
	1.00=48 or older	92	1

a. Category variable: Received surgical care (Surg_care)

b. Indicator Parameter Coding

c. Category variable: Received heart transplant (Heart_transplant)

d. Category variable: Age (median split) (AgeCat)

Block 0: Beginning Block

Omnibus Tests of Model Coefficients

-2 Log Likelihood
1401.205

Block 1: Method = Enter

Omnibus Tests of Model Coefficients[a]

-2 Log Likelihood	Overall (score)			Change From Previous Step			Change ...
	Chi-square	df	Sig.	Chi-square	df	Sig.	Chi-square
1299.938	102.244	7	.000	101.266	7	.000	101.266

Omnibus Tests of Model Coefficients[a]

Change From Previous ..	
df	Sig.
7	.000

a. Beginning Block Number 1. Method = Enter

Variables in the Equation

	B	SE	Wald	df	Sig.	Exp(B)
Year when accepted surgery	-.138	.056	6.188	1	.013	.871
Age (median split)	.801	.212	14.231	1	.000	2.227
Received surgical care	-1.015	.360	7.932	1	.005	.362
Received heart transplant	-1.247	.228	29.912	1	.000	.287
Waiting time before surgery	-.008	.004	5.692	1	.017	.992
T_COV_2	.001	.001	1.539	1	.215	1.001
T_COV_5	.001	.000	8.010	1	.005	1.001

Covariate Means

	Mean
Year when accepted surgery	70.793
Age (median split)	.418
Received surgical care	.176
Received heart transplant	.825
Waiting time before surgery	35.831
T_COV_2	34.016
T_COV_5	6170.013

```
1511  0
1512  0  *Model B (year categorised)
```

```
1513  0
1514  0  TIME PROGRAM.
1515  0  COMPUTE T_COV_2 = T_ * Age.
1516  0  COMPUTE T_COV_5 = T_ * Year_SurgeryCat.
1517  0  COXREG Survtime
1518  0    /STATUS=Status(1)
1519  0    /CONTRAST (Heart_transplant)=Indicator(1)
1520  0    /CONTRAST (Surg_care)=Indicator(1)
1521  0    /CONTRAST (Year_SurgeryCat)=Indicator(1)
1522  0    /METHOD=ENTER Year_SurgeryCat Age Surg_care Heart_transplant Wait_time T_COV_2
 T_COV_5
1523  0    /CRITERIA=PIN(.05) POUT(.10) ITERATE(20).
```

Cox Regression

Notes

Output Created		03-NOV-2026 15:35:57
Comments		
Input	Active Dataset	DataSet1
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	206
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.

Notes

Syntax		TIME PROGRAM. COMPUTE T_COV_2 = T_ * Age. COMPUTE T_COV_5 = T_ * Year_SurgeryCat. COXREG Survtime /STATUS=Status(1) /CONTRAST (Heart_transplant) =Indicator(1) /CONTRAST (Surg_care) =Indicator(1) /CONTRAST (Year_SurgeryCat) =Indicator(1) /METHOD=ENTER Year_SurgeryCat Age Surg_care Heart_transplant Wait_time T_COV_2 T_COV_5 /CRITERIA=PIN(.05) POUT(.10) ITERATE(20).
Resources	Processor Time	00:00:00.05
	Elapsed Time	00:00:00.05

Case Processing Summary

		N	Percent
Cases available in analysis	Event[a]	150	72.8%
	Censored	56	27.2%
	Total	206	100.0%
Cases dropped	Cases with missing values	0	0.0%
	Cases with negative time	0	0.0%
	Censored cases before the earliest event in a stratum	0	0.0%
	Total	0	0.0%
Total		206	100.0%

a. Dependent Variable: Survival time in days

Categorical Variable Codings[a,c,d]

		Frequency	(1)
Received surgical care[b]	0=No	182	0
	1=Yes	24	1
Received heart transplant[b]	0=No	68	0
	1=Yes	138	1
Year of surgery (median split)[b]	.00=71 or earlier	130	0
	1.00=72 or later	76	1

a. Category variable: Received surgical care (Surg_care)

b. Indicator Parameter Coding

c. Category variable: Received heart transplant (Heart_transplant)

d. Category variable: Year of surgery (median split) (Year_SurgeryCat)

Block 0: Beginning Block

Omnibus Tests of Model Coefficients

-2 Log Likelihood
1401.205

Block 1: Method = Enter

Omnibus Tests of Model Coefficients[a]

-2 Log Likelihood	Overall (score)			Change From Previous Step			Change ...
	Chi-square	df	Sig.	Chi-square	df	Sig.	Chi-square
1287.098	124.206	7	.000	114.107	7	.000	114.107

Omnibus Tests of Model Coefficients[a]

Change From Previous ..	
df	Sig.
7	.000

a. Beginning Block Number 1. Method = Enter

Variables in the Equation

	B	SE	Wald	df	Sig.	Exp(B)
Year of surgery (median split)	-.555	.262	4.479	1	.034	.574
Patient's age	.044	.011	14.730	1	.000	1.045
Received surgical care	-.697	.314	4.916	1	.027	.498
Received heart transplant	-1.382	.224	37.957	1	.000	.251
Waiting time before surgery	-.007	.004	4.445	1	.035	.993
T_COV_2	.000	.000	4.954	1	.026	1.000
T_COV_5	.004	.002	4.636	1	.031	1.004

Covariate Means

	Mean
Year of surgery (median split)	.392
Patient's age	43.961
Received surgical care	.176
Received heart transplant	.825
Waiting time before surgery	35.831
T_COV_2	3788.201
T_COV_5	26.938

```
1524  0
1525  0  *Model C (waiting time categorised)
1526  0
1527  0  TIME PROGRAM.
1528  0  COMPUTE T_COV_2 = T_ * Age.
1529  0  COMPUTE T_COV_5 = T_ * Year_surgery.
1530  0  COXREG Survtime
1531  0    /STATUS=Status(1)
1532  0    /CONTRAST (Heart_transplant)=Indicator(1)
1533  0    /CONTRAST (Surg_care)=Indicator(1)
1534  0    /CONTRAST (Wait_TimeCat)=Indicator(1)
1535  0    /METHOD=ENTER Year_surgery Age Surg_care Heart_transplant Wait_TimeCat T_COV_2
 T_COV_5
1536  0    /CRITERIA=PIN(.05) POUT(.10) ITERATE(20).
```

Cox Regression

Notes

Output Created		03-NOV-2026 15:35:57
Comments		
Input	Active Dataset	DataSet1
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	206
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
Syntax		TIME PROGRAM. COMPUTE T_COV_2 = T_ * Age. COMPUTE T_COV_5 = T_ * Year_surgery. COXREG Survtime /STATUS=Status(1) /CONTRAST (Heart_transplant) =Indicator(1) /CONTRAST (Surg_care) =Indicator(1) /CONTRAST (Wait_TimeCat)=Indicator (1) /METHOD=ENTER Year_surgery Age Surg_care Heart_transplant Wait_TimeCat T_COV_2 T_COV_5 /CRITERIA=PIN(.05) POUT(.10) ITERATE(20).
Resources	Processor Time	00:00:00.05
	Elapsed Time	00:00:00.07

Case Processing Summary

		N	Percent
Cases available in analysis	Event[a]	150	72.8%
	Censored	56	27.2%
	Total	206	100.0%
Cases dropped	Cases with missing values	0	0.0%
	Cases with negative time	0	0.0%
	Censored cases before the earliest event in a stratum	0	0.0%
	Total	0	0.0%
Total		206	100.0%

a. Dependent Variable: Survival time in days

Categorical Variable Codings[a,c,d]

		Frequency	(1)
Received surgical care[b]	0=No	182	0
	1=Yes	24	1
Received heart transplant[b]	0=No	68	0
	1=Yes	138	1
Waiting time (median split)[b]	.00=10 days or less	106	0
	1.00=More than 10 days	100	1

a. Category variable: Received surgical care (Surg_care)

b. Indicator Parameter Coding

c. Category variable: Received heart transplant (Heart_transplant)

d. Category variable: Waiting time (median split) (Wait_TimeCat)

Block 0: Beginning Block

Omnibus Tests of Model Coefficients

-2 Log Likelihood
1401.205

Block 1: Method = Enter

Omnibus Tests of Model Coefficients[a]

-2 Log Likelihood	Overall (score)			Change From Previous Step			Change ...
	Chi-square	df	Sig.	Chi-square	df	Sig.	Chi-square
1281.983	127.731	7	.000	119.222	7	.000	119.222

Omnibus Tests of Model Coefficients[a]

Change From Previous ..	
df	Sig.
7	.000

a. Beginning Block Number 1. Method = Enter

Variables in the Equation

	B	SE	Wald	df	Sig.	Exp(B)
Year when accepted surgery	-.180	.056	10.453	1	.001	.836
Patient's age	.040	.011	12.698	1	.000	1.041
Received surgical care	-.944	.358	6.963	1	.008	.389
Received heart transplant	-1.247	.256	23.690	1	.000	.287
Waiting time (median split)	-.520	.232	5.021	1	.025	.595
T_COV_2	.000	.000	6.561	1	.010	1.000
T_COV_5	.001	.000	8.344	1	.004	1.001

Covariate Means

	Mean
Year when accepted surgery	70.793
Patient's age	43.961
Received surgical care	.176
Received heart transplant	.825
Waiting time (median split)	.647
T_COV_2	3788.201
T_COV_5	6170.013

```
1537  0
1538  0
```

```
1539  0  **Comment: age categorised (median split), time term built from AgeCat. -2LL = 1
299.938, 19.1 higher than the continuous mo
           del (1280.797). T_COV_2 loses significance (p = .215). Verdict: continuous ag
e fits better.
1540  0  *Comment: year categorised (median split), -2LL = 1287.098, 6.3 higher than the
continuous model. Verdict: continuous year
           fits better.
1541  0  *Comment: waiting time categorised (median split; 106 vs 100 rows), -2LL = 1281.
983, 1.2 higher than the continuous model.
           Verdict: no gain from categorising; keep continuous.
1542  0
1543  0  *BIVARIATE SCREENING
1544  0
1545  0  *Bivariate: year with its time term.
1546  0  TIME PROGRAM.
1547  0  COMPUTE T_COV_5 = T_ * Year_surgery.
1548  0  COXREG Survtime
1549  0    /STATUS=Status(1)
1550  0    /METHOD=ENTER Year_surgery T_COV_5
1551  0    /CRITERIA=PIN(.05) POUT(.10) ITERATE(20).
```

Cox Regression

Notes

Output Created		03-NOV-2026 15:35:57
Comments		
Input	Active Dataset	DataSet1
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	206
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
Syntax		TIME PROGRAM. COMPUTE T_COV_5 = T_ * Year_surgery. COXREG Survtime /STATUS=Status(1) /METHOD=ENTER Year_surgery T_COV_5 /CRITERIA=PIN(.05) POUT(.10) ITERATE(20).
Resources	Processor Time	00:00:00.05
	Elapsed Time	00:00:00.04

Case Processing Summary

		N	Percent
Cases available in analysis	Event[a]	150	72.8%
	Censored	56	27.2%
	Total	206	100.0%
Cases dropped	Cases with missing values	0	0.0%
	Cases with negative time	0	0.0%
	Censored cases before the earliest event in a stratum	0	0.0%
	Total	0	0.0%
Total		206	100.0%

a. Dependent Variable: Survival time in days

Block 0: Beginning Block

Omnibus Tests of Model Coefficients

-2 Log Likelihood
1401.205

Block 1: Method = Enter

Omnibus Tests of Model Coefficients[a]

-2 Log Likelihood	Overall (score)			Change From Previous Step			Change ...
	Chi-square	df	Sig.	Chi-square	df	Sig.	Chi-square
1384.390	17.007	2	.000	16.815	2	.000	16.815

Omnibus Tests of Model Coefficients[a]

Change From Previous ..	
df	Sig.
2	.000

a. Beginning Block Number 1. Method = Enter

Variables in the Equation

	B	SE	Wald	df	Sig.	Exp(B)
Year when accepted surgery	-.230	.057	16.439	1	.000	.795
T_COV_5	.001	.000	3.561	1	.059	1.001

Covariate Means

	Mean
Year when accepted surgery	70.793
T_COV_5	6170.013

```
1552  0
1553  0  *Bivariate: age with its time term.
1554  0  TIME PROGRAM.
1555  0  COMPUTE T_COV_2 = T_ * Age.
1556  0  COXREG Survtime
1557  0    /STATUS=Status(1)
1558  0    /METHOD=ENTER Age T_COV_2
1559  0    /CRITERIA=PIN(.05) POUT(.10) ITERATE(20).
```

Cox Regression

Notes

Output Created		03-NOV-2026 15:35:57
Comments		
Input	Active Dataset	DataSet1
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	206
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
Syntax		TIME PROGRAM. COMPUTE T_COV_2 = T_ * Age. COXREG Survtime /STATUS=Status(1) /METHOD=ENTER Age T_COV_2 /CRITERIA=PIN(.05) POUT(.10) ITERATE(20).
Resources	Processor Time	00:00:00.03
	Elapsed Time	00:00:00.04

Case Processing Summary

		N	Percent
Cases available in analysis	Event[a]	150	72.8%
	Censored	56	27.2%
	Total	206	100.0%
Cases dropped	Cases with missing values	0	0.0%
	Cases with negative time	0	0.0%
	Censored cases before the earliest event in a stratum	0	0.0%
	Total	0	0.0%
Total		206	100.0%

a. Dependent Variable: Survival time in days

Block 0: Beginning Block

Omnibus Tests of Model Coefficients

-2 Log Likelihood
1401.205

Block 1: Method = Enter

Omnibus Tests of Model Coefficients[a]

-2 Log Likelihood	Overall (score)			Change From Previous Step			Change ...
	Chi-square	df	Sig.	Chi-square	df	Sig.	Chi-square
1386.534	13.106	2	.001	14.670	2	.001	14.670

Omnibus Tests of Model Coefficients[a]

Change From Previous ..	
df	Sig.
2	.001

a. Beginning Block Number 1. Method = Enter

Variables in the Equation

	B	SE	Wald	df	Sig.	Exp(B)
Patient's age	.020	.011	3.137	1	.077	1.020
T_COV_2	.000	.000	3.326	1	.068	1.000

Covariate Means

	Mean
Patient's age	43.961
T_COV_2	3788.201

```
1560  0
1561  0  *Bivariate: surgical care.
1562  0  COXREG Survtime
1563  0    /STATUS=Status(1)
1564  0    /CONTRAST (Surg_care)=Indicator(1)
1565  0    /METHOD=ENTER Surg_care
1566  0    /CRITERIA=PIN(.05) POUT(.10) ITERATE(20).
```

Cox Regression

Notes

Output Created		03-NOV-2026 15:35:57
Comments		
Input	Active Dataset	DataSet1
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	206
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
Syntax		COXREG Survtime /STATUS=Status(1) /CONTRAST (Surg_care) =Indicator(1) /METHOD=ENTER Surg_care /CRITERIA=PIN(.05) POUT(.10) ITERATE(20).
Resources	Processor Time	00:00:00.03
	Elapsed Time	00:00:00.03

Case Processing Summary

		N	Percent
Cases available in analysis	Event[a]	150	72.8%
	Censored	56	27.2%
	Total	206	100.0%
Cases dropped	Cases with missing values	0	0.0%
	Cases with negative time	0	0.0%
	Censored cases before the earliest event in a stratum	0	0.0%
	Total	0	0.0%
Total		206	100.0%

a. Dependent Variable: Survival time in days

Categorical Variable Codings[a]

		Frequency	(1)
Received surgical care[b]	0=No	182	0
	1=Yes	24	1

a. Category variable: Received surgical care (Surg_care)

b. Indicator Parameter Coding

Block 0: Beginning Block

Omnibus Tests of Model Coefficients

-2 Log Likelihood
1401.205

Block 1: Method = Enter

Omnibus Tests of Model Coefficients[a]

-2 Log Likelihood	Overall (score)			Change From Previous Step			Change ...
	Chi-square	df	Sig.	Chi-square	df	Sig.	Chi-square
1385.213	13.121	1	.000	15.991	1	.000	15.991

Omnibus Tests of Model Coefficients[a]

Change From Previous ..	
df	Sig.
1	.000

a. Beginning Block Number 1. Method = Enter

Variables in the Equation

	B	SE	Wald	df	Sig.	Exp(B)
Received surgical care	-1.052	.303	12.021	1	.001	.349

Covariate Means

	Mean
Received surgical care	.117

```
1567  0
1568  0  *Bivariate: heart transplant.
1569  0  COXREG Survtime
1570  0    /STATUS=Status(1)
1571  0    /CONTRAST (Heart_transplant)=Indicator(1)
1572  0    /METHOD=ENTER Heart_transplant
1573  0    /CRITERIA=PIN(.05) POUT(.10) ITERATE(20).
```

Cox Regression

Notes

Output Created		03-NOV-2026 15:35:57
Comments		
Input	Active Dataset	DataSet1
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	206
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
Syntax		COXREG Survtime /STATUS=Status(1) /CONTRAST (Heart_transplant) =Indicator(1) /METHOD=ENTER Heart_transplant /CRITERIA=PIN(.05) POUT(.10) ITERATE(20).
Resources	Processor Time	00:00:00.02
	Elapsed Time	00:00:00.06

Case Processing Summary

		N	Percent
Cases available in analysis	Event[a]	150	72.8%
	Censored	56	27.2%
	Total	206	100.0%
Cases dropped	Cases with missing values	0	0.0%
	Cases with negative time	0	0.0%
	Censored cases before the earliest event in a stratum	0	0.0%
	Total	0	0.0%
Total		206	100.0%

a. Dependent Variable: Survival time in days

Categorical Variable Codings[a]

		Frequency	(1)
Received heart transplant[b]	0=No	68	0
	1=Yes	138	1

a. Category variable: Received heart transplant (Heart_transplant)

b. Indicator Parameter Coding

Block 0: Beginning Block

Omnibus Tests of Model Coefficients

-2 Log Likelihood
1401.205

Block 1: Method = Enter

Omnibus Tests of Model Coefficients[a]

-2 Log Likelihood	Overall (score)			Change From Previous Step			Change ...
	Chi-square	df	Sig.	Chi-square	df	Sig.	Chi-square
1349.706	66.109	1	.000	51.499	1	.000	51.499

Omnibus Tests of Model Coefficients[a]

Change From Previous ..	
df	Sig.
1	.000

a. Beginning Block Number 1. Method = Enter

Variables in the Equation

	B	SE	Wald	df	Sig.	Exp(B)
Received heart transplant	-1.319	.173	58.436	1	.000	.267

Covariate Means

	Mean
Received heart transplant	.670

```
1574  0
1575  0  *Bivariate: waiting time.
1576  0  COXREG Survtime
1577  0    /STATUS=Status(1)
1578  0    /METHOD=ENTER Wait_time
1579  0    /CRITERIA=PIN(.05) POUT(.10) ITERATE(20).
```

Cox Regression

Notes

Output Created		03-NOV-2026 15:35:57
Comments		
Input	Active Dataset	DataSet1
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	206
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
Syntax		COXREG Survtime /STATUS=Status(1) /METHOD=ENTER Wait_time /CRITERIA=PIN(.05) POUT(.10) ITERATE(20).
Resources	Processor Time	00:00:00.02
	Elapsed Time	00:00:00.03

Case Processing Summary

		N	Percent
Cases available in analysis	Event[a]	150	72.8%
	Censored	56	27.2%
	Total	206	100.0%
Cases dropped	Cases with missing values	0	0.0%
	Cases with negative time	0	0.0%
	Censored cases before the earliest event in a stratum	0	0.0%
	Total	0	0.0%
Total		206	100.0%

a. Dependent Variable: Survival time in days

Block 0: Beginning Block

Omnibus Tests of Model Coefficients

-2 Log Likelihood
1401.205

Block 1: Method = Enter

Omnibus Tests of Model Coefficients[a]

-2 Log Likelihood	Overall (score)			Change From Previous Step			Change ...
	Chi-square	df	Sig.	Chi-square	df	Sig.	Chi-square
1364.759	20.081	1	.000	36.446	1	.000	36.446

Omnibus Tests of Model Coefficients[a]

Change From Previous ..	
df	Sig.
1	.000

a. Beginning Block Number 1. Method = Enter

Variables in the Equation

	B	SE	Wald	df	Sig.	Exp(B)
Waiting time before surgery	-.019	.004	23.084	1	.000	.981

Covariate Means

	Mean
Waiting time before surgery	25.738

```
1580  0
1581  0  *Comment: bivariate screening by likelihood-ratio test (cut-off p < 0.2): year w
ith T_COV_5 chi-square 16.815, 2 df, p < .0
           01;
1582  0  *age with T_COV_2 14.670, 2 df, p < .001; surgical care 15.991, 1 df, p < .001;
heart transplant 51.499, 1 df, p < .001;
1583  0  *waiting time 36.446, 1 df, p < .001. Verdict: all five variables enter the mult
ivariable model.
1584  0
```

```
1585  0  *CHECK INTERACTIONS
1586  0      *Reduced model
1587  0
1588  0  TIME PROGRAM.
1589  0  COMPUTE T_COV_2 = T_ * Age.
1590  0  COMPUTE T_COV_5 = T_ * Year_surgery.
1591  0  COXREG Survtime
1592  0    /STATUS=Status(1)
1593  0    /CONTRAST (Heart_transplant)=Indicator(1)
1594  0    /CONTRAST (Surg_care)=Indicator(1)
1595  0    /METHOD=ENTER Year_surgery Age Surg_care Heart_transplant Wait_time T_COV_2 T_
COV_5
1596  0    /CRITERIA=PIN(.05) POUT(.10) ITERATE(20).
```

Cox Regression

Notes

Output Created		03-NOV-2026 15:35:57
Comments		
Input	Active Dataset	DataSet1
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	206
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
Syntax		TIME PROGRAM. COMPUTE T_COV_2 = T_ * Age. COMPUTE T_COV_5 = T_ * Year_surgery. COXREG Survtime /STATUS=Status(1) /CONTRAST (Heart_transplant) =Indicator(1) /CONTRAST (Surg_care) =Indicator(1) /METHOD=ENTER Year_surgery Age Surg_care Heart_transplant Wait_time T_COV_2 T_COV_5 /CRITERIA=PIN(.05) POUT(.10) ITERATE(20).

Notes

Resources	Processor Time	00:00:00.05
	Elapsed Time	00:00:00.12

Case Processing Summary

		N	Percent
Cases available in analysis	Event[a]	150	72.8%
	Censored	56	27.2%
	Total	206	100.0%
Cases dropped	Cases with missing values	0	0.0%
	Cases with negative time	0	0.0%
	Censored cases before the earliest event in a stratum	0	0.0%
	Total	0	0.0%
Total		206	100.0%

a. Dependent Variable: Survival time in days

Categorical Variable Codings[a,c]

		Frequency	(1)
Received surgical care[b]	0=No	182	0
	1=Yes	24	1
Received heart transplant[b]	0=No	68	0
	1=Yes	138	1

a. Category variable: Received surgical care (Surg_care)

b. Indicator Parameter Coding

c. Category variable: Received heart transplant (Heart_transplant)

Block 0: Beginning Block

Omnibus Tests of Model Coefficients

-2 Log Likelihood
1401.205

Block 1: Method = Enter

Omnibus Tests of Model Coefficients[a]

-2 Log Likelihood	Overall (score)			Change From Previous Step			Change ...
	Chi-square	df	Sig.	Chi-square	df	Sig.	Chi-square
1280.797	127.571	7	.000	120.408	7	.000	120.408

Omnibus Tests of Model Coefficients[a]

Change From Previous ..	
df	Sig.
7	.000

a. Beginning Block Number 1. Method = Enter

Variables in the Equation

	B	SE	Wald	df	Sig.	Exp(B)
Year when accepted surgery	-.161	.055	8.487	1	.004	.851
Patient's age	.039	.011	12.401	1	.000	1.040
Received surgical care	-.989	.358	7.633	1	.006	.372
Received heart transplant	-1.377	.224	37.648	1	.000	.252
Waiting time before surgery	-.007	.003	4.312	1	.038	.993
T_COV_2	.000	.000	7.265	1	.007	1.000
T_COV_5	.001	.000	9.640	1	.002	1.001

Covariate Means

	Mean
Year when accepted surgery	70.793
Patient's age	43.961
Received surgical care	.176
Received heart transplant	.825
Waiting time before surgery	35.831
T_COV_2	3788.201
T_COV_5	6170.013

```
1597  0  *Comment: reduced model (no interactions), -2LL 1280.797, 7 df.
1598  0  *Full model
1599  0
1600  0  TIME PROGRAM.
1601  0  COMPUTE T_COV_2 = T_ * Age.
1602  0  COMPUTE T_COV_5 = T_ * Year_surgery.
1603  0  COXREG Survtime
1604  0    /STATUS=Status(1)
1605  0    /CONTRAST (Heart_transplant)=Indicator(1)
1606  0    /CONTRAST (Surg_care)=Indicator(1)
1607  0    /METHOD=ENTER Year_surgery Age Surg_care Heart_transplant Wait_time T_COV_2 T_
COV_5
1608  0       Age*Surg_care Age*Heart_transplant Age*Wait_time Age*Year_surgery
1609  0       Heart_transplant*Surg_care Surg_care*Wait_time Surg_care*Year_surgery
1610  0       Heart_transplant*Wait_time Heart_transplant*Year_surgery Wait_time*Year_sur
gery
1611  0    /CRITERIA=PIN(.05) POUT(.10) ITERATE(20).
```

Cox Regression

Notes

Output Created		03-NOV-2026 15:35:57
Comments		
Input	Active Dataset	DataSet1
	Filter	<none>
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	Split File	<none>
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Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
Syntax		TIME PROGRAM. COMPUTE T_COV_2 = T_ * Age. COMPUTE T_COV_5 = T_ * Year_surgery. COXREG Survtime /STATUS=Status(1) /CONTRAST (Heart_transplant) =Indicator(1) /CONTRAST (Surg_care) =Indicator(1) /METHOD=ENTER Year_surgery Age Surg_care Heart_transplant Wait_time T_COV_2 T_COV_5 Age*Surg_care Age*Heart_transplant Age*Wait_time Age*Year_surgery Heart_transplant*Surg_care Surg_care*Wait_time Surg_care*Year_surgery Heart_transplant*Wait_time Heart_transplant*Year_surgery Wait_time*Year_surgery /CRITERIA=PIN(.05) POUT(.10) ITERATE(20).
Resources	Processor Time	00:00:00.13
	Elapsed Time	00:00:00.17

Case Processing Summary

		N	Percent
Cases available in analysis	Event[a]	150	72.8%
	Censored	56	27.2%
	Total	206	100.0%
Cases dropped	Cases with missing values	0	0.0%
	Cases with negative time	0	0.0%
	Censored cases before the earliest event in a stratum	0	0.0%
	Total	0	0.0%
Total		206	100.0%

a. Dependent Variable: Survival time in days

Categorical Variable Codings[a,c]

		Frequency	(1)
Received surgical care[b]	0=No	182	0
	1=Yes	24	1
Received heart transplant[b]	0=No	68	0
	1=Yes	138	1

a. Category variable: Received surgical care (Surg_care)

b. Indicator Parameter Coding

c. Category variable: Received heart transplant (Heart_transplant)

Block 0: Beginning Block

Omnibus Tests of Model Coefficients

-2 Log Likelihood
1401.205

Block 1: Method = Enter

Omnibus Tests of Model Coefficients[a]

-2 Log Likelihood	Overall (score)			Change From Previous Step			Change ...
	Chi-square	df	Sig.	Chi-square	df	Sig.	Chi-square
1257.225	174.219	15	.000	143.979	15	.000	143.979

Omnibus Tests of Model Coefficients[a]

Change From Previous ..	
df	Sig.
15	.000

a. Beginning Block Number 1. Method = Enter

Variables in the Equation[b]

	B	SE	Wald	df	Sig.	Exp(B)
Year when accepted surgery	-.538	.315	2.922	1	.087	.584
Patient's age	-.382	.465	.674	1	.412	.682
Received surgical care	-60.125	27.358	4.830	1	.028	.000
Received heart transplant	7.373	8.544	.745	1	.388	1592.954
Waiting time before surgery	-.434	.181	5.734	1	.017	.648
T_COV_2	.000	.000	10.490	1	.001	1.000
T_COV_5	.001	.000	11.439	1	.001	1.001
Patient's age*Received surgical care	-.040	.082	.245	1	.621	.960
Patient's age*Received heart transplant	-.069	.028	6.213	1	.013	.933
Patient's age*Waiting time before surgery	.001	.001	2.727	1	.099	1.001
Patient's age*Year when accepted surgery	.006	.007	.858	1	.354	1.006
Received heart transplant*Received surgical care			.	0[a]	.	
Received surgical care*Waiting time before surgery	.002	.008	.090	1	.765	1.002

Variables in the Equation[b]

	B	SE	Wald	df	Sig.	Exp(B)
Received surgical care*Year when accepted surgery	.850	.367	5.355	1	.021	2.339
Received heart transplant*Waiting time before surgery			.	0[a]	.	
Received heart transplant*Year when accepted surgery	-.075	.117	.411	1	.522	.928
Waiting time before surgery*Year when accepted surgery	.005	.002	4.284	1	.038	1.005

a. Degree of freedom reduced because of constant or linearly dependent covariates

b. Constant or Linearly Dependent Covariates R = Risk set effect. Received heart transplant*Received surgical care = Received surgical care ; Received heart transplant*Waiting time before surgery = Waiting time before surgery ;

Covariate Means

	Mean
Year when accepted surgery	70.793
Patient's age	43.961
Received surgical care	.176
Received heart transplant	.825
Waiting time before surgery	35.831
T_COV_2	3788.201
T_COV_5	6170.013
Patient's age*Received surgical care	8.063
Patient's age*Received heart transplant	37.306
Patient's age*Waiting time before surgery	1625.843
Patient's age*Year when accepted surgery	3111.068

Covariate Means

	Mean
Received heart transplant*Received surgical care	.176
Received surgical care*Waiting time before surgery	7.886
Received surgical care*Year when accepted surgery	12.554
Received heart transplant*Waiting time before surgery	35.831
Received heart transplant*Year when accepted surgery	58.530
Waiting time before surgery*Year when accepted surgery	2559.188

```
1612  0  *Comment: full model, -2LL 1257.225, 15 df (SPSS dropped 2 redundant terms: Hear
t_transplant*Surg_care, Heart_transplant*Wa
           it_time).
1613  0  *LR = 1280.797 - 1257.225 = 23.57 on 8 df, above 15.51, so interactions are sign
ificant. Limitation: sample too small for a
           ll terms.
1614  0  *SELECTING INTERACTION TERMS (backward elimination, LR)
1615  0  TIME PROGRAM.
1616  0  TIME PROGRAM.
1617  0  COMPUTE T_COV_Age = T_ * Age.
1618  0  COMPUTE T_COV_YearSurg = T_ * Year_surgery.
1619  0  COXREG Survtime
1620  0    /STATUS=Status(1)
1621  0    /CONTRAST (Heart_transplant)=Indicator(1)
1622  0    /CONTRAST (Surg_care)=Indicator(1)
1623  0    /METHOD=BSTEP(LR) Year_surgery Age Surg_care Heart_transplant Wait_time T_COV_
Age T_COV_YearSurg
1624  0       Age*Surg_care Age*Heart_transplant Age*Wait_time Age*Year_surgery
1625  0       Heart_transplant*Surg_care Surg_care*Wait_time Surg_care*Year_surgery
1626  0       Heart_transplant*Wait_time Heart_transplant*Year_surgery Wait_time*Year_sur
gery
1627  0    /CRITERIA=PIN(.05) POUT(.10) ITERATE(20).
```

Cox Regression

Notes

Output Created		03-NOV-2026 15:35:57
Comments		
Input	Active Dataset	DataSet1
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	206
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
Syntax		TIME PROGRAM. COMPUTE T_COV_Age = T_ * Age. COMPUTE T_COV_YearSurg = T_ * Year_surgery. COXREG Survtime /STATUS=Status(1) /CONTRAST (Heart_transplant) =Indicator(1) /CONTRAST (Surg_care) =Indicator(1) /METHOD=BSTEP(LR) Year_surgery Age Surg_care Heart_transplant Wait_time T_COV_Age T_COV_YearSurg Age*Surg_care Age*Heart_transplant Age*Wait_time Age*Year_surgery Heart_transplant*Surg_care Surg_care*Wait_time Surg_care*Year_surgery Heart_transplant*Wait_time Heart_transplant*Year_surgery Wait_time*Year_surgery /CRITERIA=PIN(.05) POUT(.10) ITERATE(20).
Resources	Processor Time	00:00:02.67
	Elapsed Time	00:00:03.13

Case Processing Summary

		N	Percent
Cases available in analysis	Event[a]	150	72.8%
	Censored	56	27.2%
	Total	206	100.0%
Cases dropped	Cases with missing values	0	0.0%
	Cases with negative time	0	0.0%
	Censored cases before the earliest event in a stratum	0	0.0%
	Total	0	0.0%
Total		206	100.0%

a. Dependent Variable: Survival time in days

Categorical Variable Codings[a,c]

		Frequency	(1)
Received surgical care[b]	0=No	182	0
	1=Yes	24	1
Received heart transplant[b]	0=No	68	0
	1=Yes	138	1

a. Category variable: Received surgical care (Surg_care)

b. Indicator Parameter Coding

c. Category variable: Received heart transplant (Heart_transplant)

Block 0: Beginning Block

Omnibus Tests of Model Coefficients

-2 Log Likelihood
1401.205

Block 1: Method = Backward Stepwise (Likelihood Ratio)

Omnibus Tests of Model Coefficients[h]

Step	-2 Log Likelihood	Overall (score)			Change From Previous Step		
		Chi-square	df	Sig.	Chi-square	df	Sig.
1[a]	1257.225	174.219	15	.000	143.979	15	.000
4[b]	1257.315	173.204	14	.000	.090	1	.765
5[c]	1257.476	172.915	13	.000	.161	1	.689
6[d]	1257.898	167.864	12	.000	.422	1	.516
7[e]	1258.400	167.861	11	.000	.502	1	.479
8[f]	1260.239	160.193	10	.000	1.839	1	.175
9[g]	1262.727	159.750	9	.000	2.488	1	.115

Omnibus Tests of Model Coefficients[h]

Step	Change From Previous Block		
	Chi-square	df	Sig.
1[a]	143.979	15	.000
4[b]	143.890	14	.000
5[c]	143.729	13	.000
6[d]	143.307	12	.000
7[e]	142.805	11	.000
8[f]	140.966	10	.000
9[g]	138.478	9	.000

a. Variable(s) Entered at Step Number 1: Year when accepted surgery Patient's age Received surgical care Received heart transplant Waiting time before surgery T_COV_Age T_COV_YearSurg Patient's age*Received surgical care Patient's age*Received heart transplant Patient's age*Waiting

b. Variable Removed at Step Number 4: Received surgical care*Waiting time before surgery

c. Variable Removed at Step Number 5: Patient's age*Received surgical care

d. Variable Removed at Step Number 6: Received heart transplant*Year when accepted surgery

e. Variable Removed at Step Number 7: Patient's age

f. Variable Removed at Step Number 8: Received heart transplant

g. Variable Removed at Step Number 9: Patient's age*Waiting time before surgery

h. Beginning Block Number 1. Method = Backward Stepwise (Likelihood Ratio)

Variables in the Equation[b]

		B	SE	Wald	df	Sig.	Exp(B)
Step 1	Year when accepted surgery	-.538	.315	2.922	1	.087	.584
	Patient's age	-.382	.465	.674	1	.412	.682
	Received surgical care	-60.125	27.358	4.830	1	.028	.000
	Received heart transplant	7.373	8.544	.745	1	.388	1592.954
	Waiting time before surgery	-.434	.181	5.734	1	.017	.648
	T_COV_Age	.000	.000	10.490	1	.001	1.000
	T_COV_YearSurg	.001	.000	11.439	1	.001	1.001
	Patient's age*Received surgical care	-.040	.082	.245	1	.621	.960
	Patient's age*Received heart transplant	-.069	.028	6.213	1	.013	.933
	Patient's age*Waiting time before surgery	.001	.001	2.727	1	.099	1.001
	Patient's age*Year when accepted surgery	.006	.007	.858	1	.354	1.006
	Received heart transplant*Received surgical care			.	0[a]	.	
	Received surgical care*Waiting time before surgery	.002	.008	.090	1	.765	1.002
	Received surgical care*Year when accepted surgery	.850	.367	5.355	1	.021	2.339
	Received heart transplant*Waiting time before surgery			.	0[a]	.	
	Received heart transplant*Year when accepted surgery	-.075	.117	.411	1	.522	.928
	Waiting time before surgery*Year when accepted surgery	.005	.002	4.284	1	.038	1.005
Step 4	Year when accepted surgery	-.538	.315	2.916	1	.088	.584
	Patient's age	-.383	.466	.675	1	.411	.682
	Received surgical care	-62.950	26.009	5.858	1	.016	.000
	Received heart transplant	7.839	8.402	.871	1	.351	2538.681
	Waiting time before surgery	-.448	.174	6.603	1	.010	.639

Variables in the Equation[b]

		B	SE	Wald	df	Sig.	Exp(B)
	T_COV_Age	.000	.000	10.464	1	.001	1.000
	T_COV_YearSurg	.001	.000	11.393	1	.001	1.001
	Patient's age*Received surgical care	-.032	.078	.167	1	.682	.969
	Patient's age*Received heart transplant	-.070	.028	6.471	1	.011	.932
	Patient's age*Waiting time before surgery	.001	.001	3.080	1	.079	1.001
	Patient's age*Year when accepted surgery	.006	.007	.859	1	.354	1.006
	Received surgical care*Year when accepted surgery	.885	.351	6.358	1	.012	2.423
	Received heart transplant*Year when accepted surgery	-.081	.115	.496	1	.481	.922
	Waiting time before surgery*Year when accepted surgery	.005	.002	4.912	1	.027	1.005
Step 5	Year when accepted surgery	-.523	.312	2.812	1	.094	.593
	Patient's age	-.361	.461	.612	1	.434	.697

Variables in the Equation[b]

		B	SE	Wald	df	Sig.	Exp(B)
	Received surgical care	-65.802	25.759	6.526	1	.011	.000
	Received heart transplant	7.273	8.279	.772	1	.380	1441.245
	Waiting time before surgery	-.429	.168	6.495	1	.011	.651
	T_COV_Age	.000	.000	10.380	1	.001	1.000
	T_COV_YearSurg	.001	.000	11.317	1	.001	1.001
	Patient's age*Received heart transplant	-.069	.027	6.340	1	.012	.934
	Patient's age*Waiting time before surgery	.001	.001	3.020	1	.082	1.001
	Patient's age*Year when accepted surgery	.006	.007	.790	1	.374	1.006
	Received surgical care*Year when accepted surgery	.904	.358	6.364	1	.012	2.469
	Received heart transplant*Year when accepted surgery	-.074	.114	.425	1	.515	.929
	Waiting time before surgery*Year when accepted surgery	.005	.002	4.727	1	.030	1.005
Step 6	Year when accepted surgery	-.521	.311	2.799	1	.094	.594
	Patient's age	-.322	.457	.497	1	.481	.725

Variables in the Equation[b]

		B	SE	Wald	df	Sig.	Exp(B)
	Received surgical care	-65.081	25.491	6.518	1	.011	.000
	Received heart transplant	1.939	1.295	2.242	1	.134	6.953
	Waiting time before surgery	-.374	.146	6.610	1	.010	.688
	T_COV_Age	.000	.000	10.331	1	.001	1.000
	T_COV_YearSurg	.001	.000	10.900	1	.001	1.001
	Patient's age*Received heart transplant	-.066	.027	5.907	1	.015	.936
	Patient's age*Waiting time before surgery	.001	.001	2.959	1	.085	1.001
	Patient's age*Year when accepted surgery	.005	.007	.661	1	.416	1.005
	Received surgical care*Year when accepted surgery	.894	.355	6.354	1	.012	2.444
	Waiting time before surgery*Year when accepted surgery	.004	.002	4.579	1	.032	1.004
Step 7	Year when accepted surgery	-.306	.063	23.317	1	.000	.736
	Received surgical care	-64.017	25.213	6.447	1	.011	.000
	Received heart transplant	1.605	1.172	1.875	1	.171	4.979
	Waiting time before surgery	-.390	.143	7.396	1	.007	.677
	T_COV_Age	.000	.000	10.176	1	.001	1.000
	T_COV_YearSurg	.001	.000	11.517	1	.001	1.001
	Patient's age*Received heart transplant	-.059	.025	5.766	1	.016	.942
	Patient's age*Waiting time before surgery	.001	.001	3.532	1	.060	1.001
	Patient's age*Year when accepted surgery	.001	.000	13.306	1	.000	1.001
	Received surgical care*Year when accepted surgery	.879	.351	6.282	1	.012	2.408
	Waiting time before surgery*Year when accepted surgery	.004	.002	5.108	1	.024	1.004
Step 8	Year when accepted surgery	-.287	.062	21.666	1	.000	.751
	Received surgical care	-67.952	24.902	7.446	1	.006	.000

Variables in the Equation[b]

		B	SE	Wald	df	Sig.	Exp(B)
	Waiting time before surgery	-.338	.137	6.035	1	.014	.713
	T_COV_Age	.000	.000	9.916	1	.002	1.000
	T_COV_YearSurg	.001	.000	11.589	1	.001	1.001
	Patient's age*Received heart transplant	-.026	.005	29.002	1	.000	.974
	Patient's age*Waiting time before surgery	.001	.001	2.187	1	.139	1.001
	Patient's age*Year when accepted surgery	.001	.000	12.342	1	.000	1.001
	Received surgical care*Year when accepted surgery	.934	.346	7.286	1	.007	2.546
	Waiting time before surgery*Year when accepted surgery	.004	.002	4.580	1	.032	1.004
Step 9	Year when accepted surgery	-.294	.061	23.161	1	.000	.745
	Received surgical care	-72.292	23.661	9.335	1	.002	.000
	Waiting time before surgery	-.299	.136	4.844	1	.028	.742
	T_COV_Age	.000	.000	10.499	1	.001	1.000
	T_COV_YearSurg	.001	.000	10.864	1	.001	1.001
	Patient's age*Received heart transplant	-.027	.005	29.601	1	.000	.974
	Patient's age*Year when accepted surgery	.001	.000	18.434	1	.000	1.001
	Received surgical care*Year when accepted surgery	.996	.329	9.178	1	.002	2.707
	Waiting time before surgery*Year when accepted surgery	.004	.002	4.627	1	.031	1.004

a. Degree of freedom reduced because of constant or linearly dependent covariates

b. Constant or Linearly Dependent Covariates R = Risk set effect. Received heart transplant*Received surgical care = Received surgical care ; Received heart transplant*Waiting time before surgery = Waiting time before surgery ;

Model if Term Removed

Term Removed		Loss Chi-square	df	Sig.
Step 3	Year when accepted surgery	3.011	1	.083
	Patient's age	.684	1	.408
	Received surgical care	5.252	1	.022
	Received heart transplant	.738	1	.390
	Waiting time before surgery	5.557	1	.018
	T_COV_Age	13.757	1	.000
	T_COV_YearSurg	11.293	1	.001
	Patient's age*Received surgical care	.231	1	.631
	Patient's age*Received heart transplant	6.424	1	.011
	Patient's age*Waiting time before surgery	3.079	1	.079
	Patient's age*Year when accepted surgery	.873	1	.350
	Received surgical care*Waiting time before surgery	.090	1	.765
	Received surgical care*Year when accepted surgery	5.554	1	.018
	Received heart transplant*Year when accepted surgery	.409	1	.523
	Waiting time before surgery*Year when accepted surgery	4.166	1	.041
Step 4	Year when accepted surgery	3.006	1	.083
	Patient's age	.684	1	.408
	Received surgical care	6.319	1	.012
	Received heart transplant	.861	1	.353
	Waiting time before surgery	6.698	1	.010
	T_COV_Age	13.668	1	.000
	T_COV_YearSurg	11.259	1	.001
	Patient's age*Received surgical care	.161	1	.689

Model if Term Removed

Term Removed		Loss Chi-square	df	Sig.
	Patient's age*Received heart transplant	6.718	1	.010
	Patient's age*Waiting time before surgery	3.612	1	.057
	Patient's age*Year when accepted surgery	.874	1	.350
	Received surgical care*Year when accepted surgery	6.574	1	.010
	Received heart transplant*Year when accepted surgery	.493	1	.483
	Waiting time before surgery*Year when accepted surgery	4.984	1	.026
Step 5	Year when accepted surgery	2.890	1	.089
	Patient's age	.620	1	.431
	Received surgical care	6.757	1	.009
	Received heart transplant	.763	1	.382
	Waiting time before surgery	6.663	1	.010
	T_COV_Age	13.530	1	.000
	T_COV_YearSurg	11.195	1	.001
	Patient's age*Received heart transplant	6.559	1	.010
	Patient's age*Waiting time before surgery	3.508	1	.061
	Patient's age*Year when accepted surgery	.803	1	.370
	Received surgical care*Year when accepted surgery	6.539	1	.011
	Received heart transplant*Year when accepted surgery	.422	1	.516
	Waiting time before surgery*Year when accepted surgery	4.836	1	.028
Step 6	Year when accepted surgery	2.873	1	.090
	Patient's age	.502	1	.479

Model if Term Removed

Term Removed		Loss Chi-square	df	Sig.
	Received surgical care	6.719	1	.010
	Received heart transplant	2.285	1	.131
	Waiting time before surgery	6.870	1	.009
	T_COV_Age	13.474	1	.000
	T_COV_YearSurg	10.773	1	.001
	Patient's age*Received heart transplant	6.190	1	.013
	Patient's age*Waiting time before surgery	3.438	1	.064
	Patient's age*Year when accepted surgery	.670	1	.413
	Received surgical care*Year when accepted surgery	6.499	1	.011
	Waiting time before surgery*Year when accepted surgery	4.774	1	.029
Step 7	Year when accepted surgery	24.106	1	.000
	Received surgical care	6.599	1	.010
	Received heart transplant	1.839	1	.175
	Waiting time before surgery	7.750	1	.005
	T_COV_Age	13.268	1	.000
	T_COV_YearSurg	11.382	1	.001
	Patient's age*Received heart transplant	5.691	1	.017
	Patient's age*Waiting time before surgery	4.309	1	.038
	Patient's age*Year when accepted surgery	15.662	1	.000
	Received surgical care*Year when accepted surgery	6.380	1	.012
	Waiting time before surgery*Year when accepted surgery	5.371	1	.020
Step 8	Year when accepted surgery	22.316	1	.000
	Received surgical care	7.558	1	.006

Model if Term Removed

Term Removed		Loss Chi-square	df	Sig.
	Waiting time before surgery	6.284	1	.012
	T_COV_Age	12.596	1	.000
	T_COV_YearSurg	11.436	1	.001
	Patient's age*Received heart transplant	28.014	1	.000
	Patient's age*Waiting time before surgery	2.488	1	.115
	Patient's age*Year when accepted surgery	13.860	1	.000
	Received surgical care*Year when accepted surgery	7.334	1	.007
	Waiting time before surgery*Year when accepted surgery	4.745	1	.029
Step 9	Year when accepted surgery	23.998	1	.000
	Received surgical care	9.206	1	.002
	Waiting time before surgery	5.013	1	.025
	T_COV_Age	13.330	1	.000
	T_COV_YearSurg	10.696	1	.001
	Patient's age*Received heart transplant	28.685	1	.000
	Patient's age*Year when accepted surgery	20.865	1	.000
	Received surgical care*Year when accepted surgery	8.966	1	.003
	Waiting time before surgery*Year when accepted surgery	4.752	1	.029

Variables not in the Equation[a,b,c,d,e,f]

		Score	df	Sig.
Step 4	Received heart transplant*Received surgical care	.	0	.
	Received surgical care*Waiting time before surgery	.090	1	.764
	Received heart transplant*Waiting time before surgery	.	0	.
Step 5	Patient's age*Received surgical care	.168	1	.682
	Received heart transplant*Received surgical care	.	0	.
	Received surgical care*Waiting time before surgery	.019	1	.890
	Received heart transplant*Waiting time before surgery	.	0	.
Step 6	Patient's age*Received surgical care	.092	1	.761
	Received heart transplant*Received surgical care	.	0	.
	Received surgical care*Waiting time before surgery	.077	1	.782
	Received heart transplant*Waiting time before surgery	.	0	.
	Received heart transplant*Year when accepted surgery	.425	1	.514
Step 7	Patient's age	.497	1	.481
	Patient's age*Received surgical care	.057	1	.812
	Received heart transplant*Received surgical care	.	0	.
	Received surgical care*Waiting time before surgery	.079	1	.778

Variables not in the Equation[a,b,c,d,e,f]

		Score	df	Sig.
	Received heart transplant*Waiting time before surgery	.	0	.
	Received heart transplant*Year when accepted surgery	.305	1	.581
Step 8	Patient's age	.056	1	.812
	Received heart transplant	1.880	1	.170
	Patient's age*Received surgical care	.012	1	.912
	Received heart transplant*Received surgical care	.	0	.
	Received surgical care*Waiting time before surgery	.183	1	.668
	Received heart transplant*Waiting time before surgery	.	0	.
	Received heart transplant*Year when accepted surgery	1.622	1	.203
Step 9	Patient's age	.897	1	.344
	Received heart transplant	.018	1	.893
	Patient's age*Received surgical care	.215	1	.643
	Patient's age*Waiting time before surgery	2.240	1	.134
	Received heart transplant*Received surgical care	.	0	.
	Received surgical care*Waiting time before surgery	1.026	1	.311
	Received heart transplant*Waiting time before surgery	.	0	.
	Received heart transplant*Year when accepted surgery	.008	1	.930

a. Residual Chi Square = .090 with 1 df Sig. = .764

b. Residual Chi Square = .266 with 2 df Sig. = .875

c. Residual Chi Square = .688 with 3 df Sig. = .876

d. Residual Chi Square = 1.182 with 4 df Sig. = .881

e. Residual Chi Square = 3.025 with 5 df Sig. = .696

f. Residual Chi Square = 4.881 with 6 df Sig. = .559

Covariate Means

	Mean
Year when accepted surgery	70.793
Patient's age	43.961
Received surgical care	.176
Received heart transplant	.825
Waiting time before surgery	35.831
T_COV_Age	3788.201
T_COV_YearSurg	6170.013
Patient's age*Received surgical care	8.063
Patient's age*Received heart transplant	37.306
Patient's age*Waiting time before surgery	1625.843
Patient's age*Year when accepted surgery	3111.068
Received heart transplant*Received surgical care	.176
Received surgical care*Waiting time before surgery	7.886
Received surgical care*Year when accepted surgery	12.554
Received heart transplant*Waiting time before surgery	35.831
Received heart transplant*Year when accepted surgery	58.530
Waiting time before surgery*Year when accepted surgery	2559.188